



SERINGAL RIO BRANCO REDD+ PROJECT

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Prepared by	<i>Biofíllica Ambipar Environmental Investments S.A</i>

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Seringal Rio Branco REDD+ Project is a partnership between Biofíllica Ambipar Environmental Investments S.A and Amazônia Agroindústria EIRELI, with the aim of promoting forest conservation and mitigating climate change by reducing Greenhouse Gas (GHG) emissions arising from changes in land use. In this sense, it is expected that throughout its life cycle, the Project will avoid the total emission of 3.247.296 tCO₂eq for the 40 years of its validity. In addition to climate benefits, the Project also aims to generate socio-environmental benefits based on sustainable economic development practices and improving the well-being of surrounding communities, while preserving the culture and tradition of local people.

The Seringal Rio Branco REDD+ Project is in the Alto Juruá region, between the municipalities of Cruzeiro do Sul and Porto Walter, which covers an area of 29,616 hectares. The Juruá River Basin is formed by four micro-basins: the Igarapé do Meio micro-basin, the Igarapé Primeiro de Mar micro-basin and the Igarapé Grande micro-basin, in addition to the Valparaíso River basin itself (micro-basin), which is the main drainage network of the rubber plantation region.

The project aims to generate reductions in GHG emissions through activities aimed at conserving the forest and its natural resources, as well as maintaining carbon stocks, through activities that promote the reduction of deforestation in the region, such as improving heritage surveillance, monitoring changes in land use and land cover using satellite images and strengthening the management of non-timber forest products. The development of activities combined with good project management, with continuous monitoring and evaluation of activities and results, will allow the Project to achieve the expected objectives and impacts.

Considering the scenario of the region that existed before the implementation of the Project, it was identified that the main vectors of deforestation and degradation that operate in the Reference Region during the historical period are: creation of settlement projects with the opening of new branch lines, conversion of areas for beef cattle raising, opening of primary forest areas for the implementation of annual clearings and opening of roads for selective wood collection. These factors contributed to the degradation process and threatened the forest area, increasing environmental impacts in the region.

Therefore, it is expected that the project, through curbing deforestation, will average approximately 81.182 tCO₂e in annual GHG emissions reductions and 3.247.296 tCO₂e in GHG emissions reductions over the 40 years of the project. All activities proposed to contain deforestation will become economically viable by combining the management of non-timber forest products together with the commercialization of carbon credits through REDD+ mechanisms.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/verification	2024	VCS	RINA	N/A

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	N/A
Project activity type	N/A

Sectoral scope	Scope 14 – Agriculture, Forestry and Other Land Uses (AFOLU);
AFOLU project category ²	Reducing Emissions from Deforestation and Forest Degradation (REDD);
Project activity type	Avoided Planned Deforestation (APD) and Avoided Unplanned Deforestation (AUD)

1.4 Project Eligibility

1.4.1 General eligibility

Pipeline listing date	26-March-2024
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¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

² See Appendix 1 of the VCS Standard

Opening meeting date	N/A
Validation date	N/A

The project falls within Sectoral Scope 14, focusing on Agriculture, Forestry, and Other Land Uses (AFOLU). It is categorized under Reducing Emissions from Deforestation and Forest Degradation (REDD). Activities primarily involve preventing Planned Deforestation (APD) and Unplanned Deforestation (AUD). This project is not grouped with others.

1.4.2 AFOLU project eligibility

Under the parameters of the VCS Methodology Requirements, v 4.5, the scope of avoided unplanned deforestation and/or degradation (AUDD) and Avoided Planned Deforestation (APD) has the Project eligibility as a premise and must include activities to reduce greenhouse gas (GHG) emissions by avoiding deforestation and/or forest degradation. Therefore, Seringal Rio Branco REDD+ Project proposes actions to reduce (GHG) emissions with activities to contain unplanned and planned deforestation and forest degradation.

Furthermore, according to decision 11/CP.7 of the Marrakesh Agreement, forests are defined as: "area of at least 0.05-1.0 ha with canopy cover (or equivalent density) of more than 10-30%, with trees with possibility to reach a minimum height of 2-5 meters at maturity in situ. A forest may consist of both closed (dense) forest formations, where trees of various strata and shelterwood cover a high proportion of the ground, and open forests. Recent natural population settlements and all plantations that have yet to reach a density of 10-30% and a height between 2 and 5 meters are included as forest, as well as areas that are normally part of the forest area and are temporarily deforested as a result of human intervention, such as harvesting, or natural causes, but where forest reversion is expected (UNFCCC, 2002)". As such, the Project Area meets the UNFCCC definition of forests and qualifies as forest for at least 10 years prior to the Project start date.

Other VCS eligibility requirements that the project meets relate to:

- The project applies a methodology within the VCS Program;
- The implementation of the project activities does not violate any applicable law;
- The project is not covered by a REDD+ jurisdictional program;

- The project will not be implemented in wetlands and does not drain native ecosystems or degrade hydrological functions;
- The risk of non-permanence will be analyzed according to the VCS Program (AFOLU Non-Permanence Risk Tool).

1.4.3 Transfer project eligibility

Not applicable.

1.5 Project Design

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

The project is not qualified as a grouped project.

1.6 Project Proponent

Organization name	Biofílica Ambipar Environmental Investments S.A
Contact person	Plínio Ribeiro
Title	CEO Executive Director
Address	Rua Vieira de Moraes, 420 – Cj. 43/44 – Campo Belo CEP: 04617-000, São Paulo/SP – Brasil
Telephone	+55 11 3073-0430
Email	plinio@biofilica.com.br

Organization name	Amazônia Agroindústria EIRELI
Contact person	James Castro Cameli
Title	Managing partner

Address	Estrada do Canela Fina, bairro Boca da Alemanha, nº 1226 CEP: 69.980-000, Cruzeiro do Sul - Acre
Telephone	+55 68 99926-1691
Email	jamescameli@yahoo.com.br

1.7 Other Entities Involved in the Project

Organization name	Vasta Insumos da Amazônia Ltda
Role in the project	
Contact person	James Castro Cameli
Title	Managing partner
Address	Rua Pedro Teles nº 360 Bairro: Centro, Cruzeiro do Sul, Acre
Telephone	+55 68 99926-1691
Email	jamescameli@yahoo.com.br

Organization name	STA Soluções Técnicas e Serviços de Engenharia Ambiental Ltda.
Role in the project	Developer of diagnostics
Contact person	Raniery Branco
Title	Owner partner
Address	Rua 28 de setembro, 1226 – Reduto ZIP 66053- 355, Belém/PA – Brasil
Telephone	+55 91 3355-4408 & +55 91 9161-2698
Email	raniery@stambiental.com

1.8 Ownership

Amazônia Agroindústria EIRELI is the legal owner of the property where the Seringal Rio Branco

REDD+ Project is being implemented and developed. The use and ownership rights are demonstrated by the following documents, which were forwarded to the audit team:

- Full Certificate containing the description of the Enrollment;
- Environmental Rural Registry (CAR);
- Certificate of Registration of Rural Property (CCIR);
- Descriptive memorial and plan (part I and II) linked to the Land Management System (SIGEF);

Based on these information and documents, it is demonstrated that the right to use the Project Area is observed according to the criteria of VCS Standard v4.3 (page 24):

"1) Right to use resulting from or granted under statute, regulation or decree by a relevant authority.

2) Right to use arising from law.

4) Right to use arising from a statutory, property, or contractual right to land, vegetation, or conservation process, or management allowing reduction of GHG emissions and/or removals (where such right includes the right to use such reductions or removals and the project proponent has not been divested of such right to use)."

In a supplementary manner, by signing a contract between the two bidders, responsibilities and rights over the Project were set out, as well as the percentage of carbon credits addressed to each party.

1.9 Project Start Date

Project start date	18-March-2023
Justification	The project start date is compliant with the VCS Program requirements.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☐ Ten years, fixed ☒ Other (40 years)
Start and end date of first or fixed crediting period	18-March-2023 to 17-March-2063

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☒ < 300,000 tCO₂e/year (project)
- ☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
18-Mar-2023 to 17-Mar-2024	181.673
18-Mar-2024 to 17-Mar-2025	183.549
18-Mar-2025 to 17-Mar-2026	185.425
18-Mar-2026 to 17-Mar-2027	197.040
18-Mar-2027 to 17-Mar-2028	199.014
18-Mar-2028 to 17-Mar-2029	200.989
18-Mar-2029 to 17-Mar-2030	202.964
18-Mar-2030 to 17-Mar-2031	215.117
18-Mar-2031 to 17-Mar-2032	217.194
18-Mar-2032 to 17-Mar-2033	219.270
18-Mar-2033 to 17-Mar-2034	26.299
18-Mar-2034 to 17-Mar-2035	30.322
18-Mar-2035 to 17-Mar-2036	28.471
18-Mar-2036 to 17-Mar-2037	26.499
18-Mar-2037 to 17-Mar-2038	30.592

18-Mar-2038 to 17-Mar-2039	28.697
18-Mar-2039 to 17-Mar-2040	26.802
18-Mar-2040 to 17-Mar-2041	24.785
18-Mar-2041 to 17-Mar-2042	28.903
18-Mar-2042 to 17-Mar-2043	26.965
18-Mar-2043 to 17-Mar-2044	29.623
18-Mar-2044 to 17-Mar-2045	29.671
18-Mar-2045 to 17-Mar-2046	36.121
18-Mar-2046 to 17-Mar-2047	36.297
18-Mar-2047 to 17-Mar-2048	36.345
18-Mar-2048 to 17-Mar-2049	36.462
18-Mar-2049 to 17-Mar-2050	43.172
18-Mar-2050 to 17-Mar-2051	43.351
18-Mar-2051 to 17-Mar-2052	43.400
18-Mar-2052 to 17-Mar-2053	50.367
18-Mar-2053 to 17-Mar-2054	50.550
18-Mar-2054 to 17-Mar-2055	50.733
18-Mar-2055 to 17-Mar-2056	50.650

18-Mar-2056 to 17-Mar-2057	58.010
18-Mar-2057 to 17-Mar-2058	58.198
18-Mar-2058 to 17-Mar-2059	58.385
18-Mar-2059 to 17-Mar-2060	58.436
18-Mar-2060 to 17-Mar-2061	65.923
18-Mar-2061 to 17-Mar-2062	66.114
18-Mar-2062 to 17-Mar-2063	64.918
Total estimated ERRs during the first or fixed crediting period	3.247.296
Total number of years	40
Average annual ERRs	81.182

1.12 Description of the Project Activity

All activities designed for the Project and their positive impacts in the short, medium and long term were carefully designed and planned, in addition to considering in this planning the need for the results to become self-sustainable in the long term.

That said, in order to maintain and improve the benefits for the climate, community and biodiversity during the 40 years of validity and beyond, several Project activities are focused on improving local governance, management, education, promotion, decision-making and greater awareness and capacity for sustainable resource management for all interested parties. These activities will have short, medium and long-term impacts and will therefore help to empower and guide all stakeholders in self-determining sustainable pathways to achieve climate benefits, community and biodiversity well beyond the life of the Project.

The strategies associated with each activity so that benefits occur during the Project life cycle and beyond are:

- Improvement in property surveillance procedures: by providing additional tools such as remote monitoring using high-resolution satellite images and offering training to the property surveillance team,

the Project aims to increase the efficiency of property surveillance operations. In this way, surveillance operations will have a great increase in the intelligence process related to monitoring and territorial management, which should directly reflect on the maintenance of climate benefits beyond the life of the Project;

- Non-timber Forest Management in Seringal Rio Branco: with knowledge of the main opportunities aimed at improving the best use of non-timber forest products, in addition to training promotions and other actions within the mapped potential lines, the Project aims to increase the clarification of the parties interested parties about the importance of natural resources and ecosystem services, valuing sustainable practices, in addition to strengthening territorial ties and, consequently, reducing rural exodus. Thus, the project must guarantee the long-term maintenance of the benefits generated, based on the empowerment of interested parties regarding the extraction of non-timber forest products, income generation and, consequently, mitigation and prevention of deforestation and forest degradation;

- Promotion of sustainable practices and environmental education related to hunting and fishing: through the mapping of opportunities for development and improvement of sustainable practices, the promotion of training and other assessments within the mapped potential lines, as well as the development of actions of environmental education related to hunting and fishing, the Project aims in the long term to promote the socioeconomic development of the region, empower communities, reduce the impacts generated by hunting and fishing actions and, consequently, knowledge about these themes , promoting mitigation and prevention of deforestation and unsustainable activities, in addition to the consequent reduction of emissions and pressure on local biodiversity. Therefore, the skills, training and educational activities that will be behind these activities will be learned by stakeholders throughout the life of the Project and will be maintained, providing lessons learned and greater enrichment of learning beyond the life of the Project;

- Biodiversity Monitoring: the Project's axis of action is the implementation of biodiversity monitoring in the Project Area throughout its life cycle, but, going forward, it will seek to provide feedback and improve the engagement of interested parties in this activity, seeking environmental involvement and awareness. In this way, it is expected that the benefits associated with these actions will help, in some way, to change the view of interested parties on local biodiversity, providing greater knowledge to people and, thus, reducing the impacts caused between human-human conflict. nature beyond the lifetime of the Project.

- Promotion of infrastructure insertion and maintenance actions: the Project proposes strategic actions, such as mapping development opportunities in partnership with communities and the Public Authorities, aiming to strengthen existing practices and implement infrastructural improvements. The identification of interested parties and the implementation of partnerships add strength and financial resources to the initiatives, consolidating an integrated and sustainable approach to the progress of riverside communities.

1.13 Project Location

The State of Acre is located in the extreme southwest of the Brazilian Amazon, between latitudes 07°07'S and 11°08'S, and longitudes 66°30'W and 74°WGr. Its territorial surface is 164,221.36 km2, which corresponds to 4% of the Brazilian Amazon and 1.9% of Brazil

The State of Acre has 22 municipalities, which, for better coordination of the state administration, were grouped into five development regions: Alto and Baixo Acre, Purus, TarauacáEnvira and Juruá, taking the main river basins as a reference.

The Seringal Rio Branco REDD+ Project, whose area is 29,616 hectares, is located within the municipal limits of Cruzeiro do Sul, while its reference region (with an area of 739,065.43 ha) covers the municipal limits of Mâncio Lima, Rodrigues Alves, Cruzeiro do Sul and Porto Walter (Figure 1).

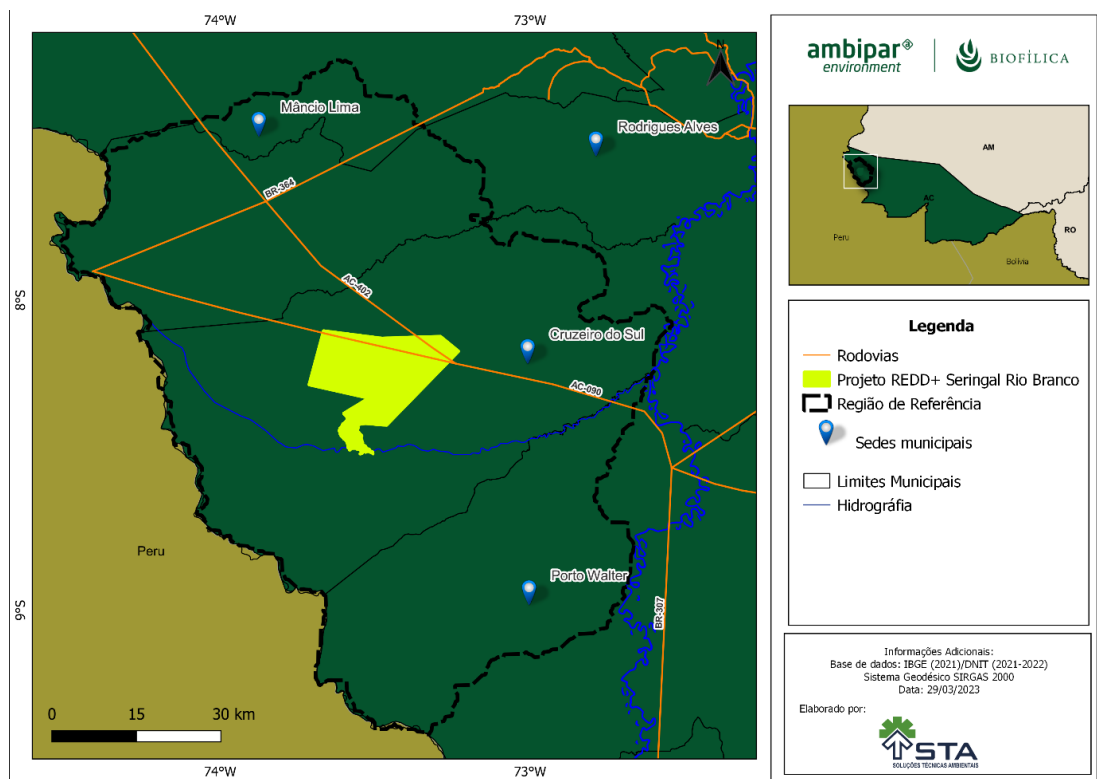


Figure 1 - Location of the Seringal Rio Branco REDD+ Project reference region.

1.14 Conditions Prior to Project Initiation

The State of Acre is located in the extreme southwest of the Brazilian Amazon, between latitudes 07°07'S and 11°08'S, and longitudes 66°30'W and 74°WGr. Its territorial surface is 164,221.36

km², which corresponds to 4% of the Brazilian Amazon and 1.9% of Brazil. In the north-south direction, the territorial extension of Acre is 445 km, being greater in the east-west direction, with 809 km. The state has a privileged geopolitical position, bordering internationally with the Department of Pando (Bolivia) and Madre de Dios (Peru) and nationally with the states of Amazonas and Rondônia, being considered the gateway to the Western Amazon (EMBRAPA, 2012).

The State of Acre has 22 municipalities, which, for better coordination of the state administration, were grouped into five development regions: Alto and Baixo Acre, Purus, TarauacáEnvira and Juruá, taking the main river basins as a reference.

The Seringal Rio Branco REDD+ Project, whose area is 29,616 hectares, is located within the municipal limits of Cruzeiro do Sul, while its reference region (with an area of 739,065.43 ha) covers the municipal limits of Mâncio Lima, Rodrigues Alves, Cruzeiro do Sul and Porto Walter.

To characterize the physical parameters of the region and the project area, hydrographic, climatic, geological, geomorphological, pedological and mineral resource data were collected, which will be detailed below:

Climate Aspects:

The Köppen-Giger International Classification for climate types is mainly based on the amount and annual distribution of precipitation, in addition to the monthly, annual and average temperature values of the coldest month. The Amazon climate was classified by Köppen as type A, that is, humid tropical, with the average temperature of the coldest month never lower than 18°C.

Acre's climate is equatorial (hot and humid), based on the Köppen classification, with two well-defined seasons, one dry and one rainy. In this sense, the climate type is Af (equatorial or humid tropical), with the occurrence of a short-term dry season and high rainfall in the rainy season (SEMA, 2019)³. Likewise, according to the database provided by Alvares et al. (2013)⁴ in the reference region, as well as in the Seringal Rio Branco REDD+ Project area, there is a predominance of the Af climate type.

The Seringal Rio Branco REDD+ project area is in an area where there is a predominance of an average temperature that fluctuates very little (25.29 °C to 25.35 °C), which represents greater viability for project installations that may require environments with constant annual meteorological variable behavior, considering, however, seasonality during the year. For the reference region, the months October to November have the highest temperatures, while the months of May to August are the least hot months.

³ Contextualização ambiental, geográfica e socioeconômica do complexo de florestas estaduais do rio gregório e seu entorno. Estado do acre. Secretaria de estado de meio ambiente – sema. Edital da concorrência nº xx/2020 – anexo 3. 2019.

⁴ ALVARES, C. A.; STAPE, J. L.; SENTELHAS, P. C.; GONÇALVES, J. L. M.; SPAROVEK, G. Köppen's climate classification map for Brazil. Meteorologische Zeitschrift, v. 22, n. 6, p. 711–728, 2013.

To characterize rainfall in the reference region, a survey was carried out in the CHIRPS Product database, developed by the United States Geological Survey (USGS) and the Climate Hazards Group at the University of California, Santa Barbara (UCSB) (Funk et al., 2015)⁵.

Thus, the average rainfall accumulated in the reference region of the Seringal Rio Branco REDD+ Project ranged from 70.73 mm to 271.27 mm. The rainiest months are from January to April, with the maximum occurring in March (Figure 2).

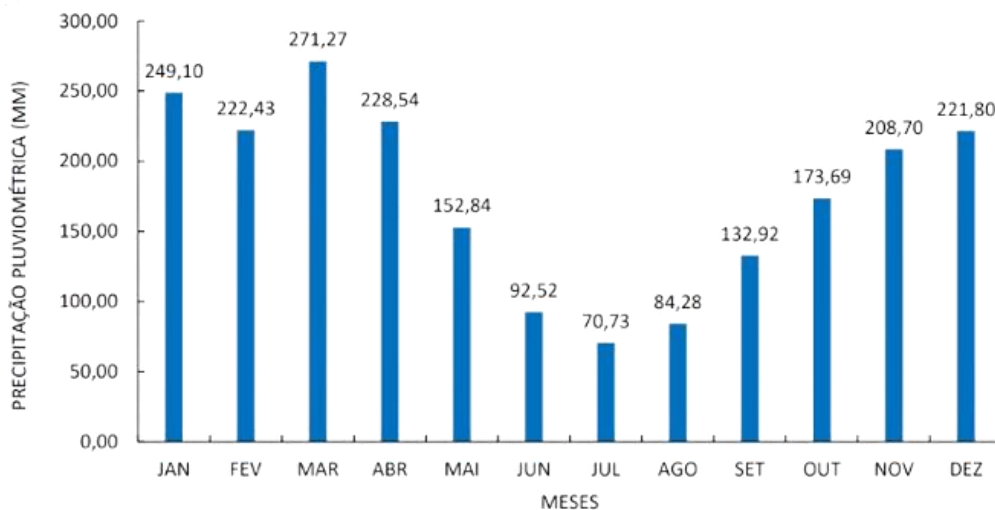


Figure 2 - Average accumulated precipitation in the reference region of the Seringal Rio Branco REDD+ Project. Source: (2023).

For the REDD+ project area, precipitation varies from 173.87 mm to 179.06 mm, with a variation of approximately 6 mm (Figure 3)

⁵ FUNK, C., PETERSON, P., LANDSFELD, M., PEDREROS, D., VERDIN, J., SHUKLA, S., HUSAK, G., ROWLAND, J., HARRISON, L., HOELL, A., MICHAELSEN, J., 2015. The climate hazards infrared precipitation with Stations a new environmental record for monitoring extremes. Scientific data 2, 1-21

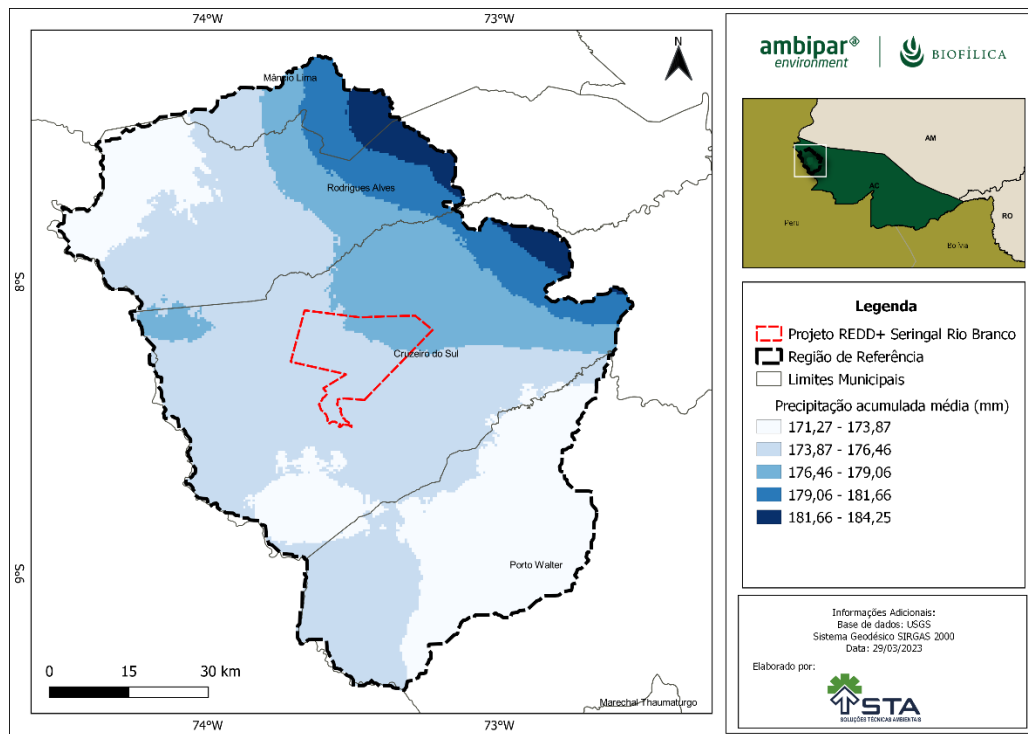


Figure 3 - Spatialization of precipitation in the reference region of the Seringal Rio Branco REDD+ Project.

Survey of the average accumulated wind speed in the Project reference region. To this end, data from the WorldClim version 2.1 database were used for the period from 1970 to 2000. Based on the results, it was observed that the lowest wind speeds were recorded in the months of April to August, with the lowest speed in the month of April (2.27 km/h) and the months of November, December and January presented the highest speeds, with a maximum in December (3.37 km/h), that is, the increase in speeds occurs during the rainy season, especially in the months of November, December and January, which may be related to the formation of clouds and the occurrence of rainfall itself.

Hydrographic Aspects:

Acre's hydrography makes up the large Amazon hydrographic basin and the state's main watercourses run in a southwest-northeast direction, practically parallel from east to west (Figure). It is formed by the Juruá and Purus River basins, tributaries of the right bank of the Solimões River. The main rivers that cross it have their sources located in Peru, cross the state and flow into rivers in the territory of the state of Amazonas. These are tributaries on the right bank of the Solimões River, which, from Manaus, is called Amazonas. They are meandering rivers, a typical form of lowland rivers. The main watercourses are the Tarauacá, Purus, Gregório, Envira, Acre and Juruá rivers. They form the state hydrographic network, divided between the Acre-Purus Basin and the Juruá Basin. The state's rivers constitute a very important means of transport, so much so that most of Acre's cities and towns are built on the banks of the rivers.

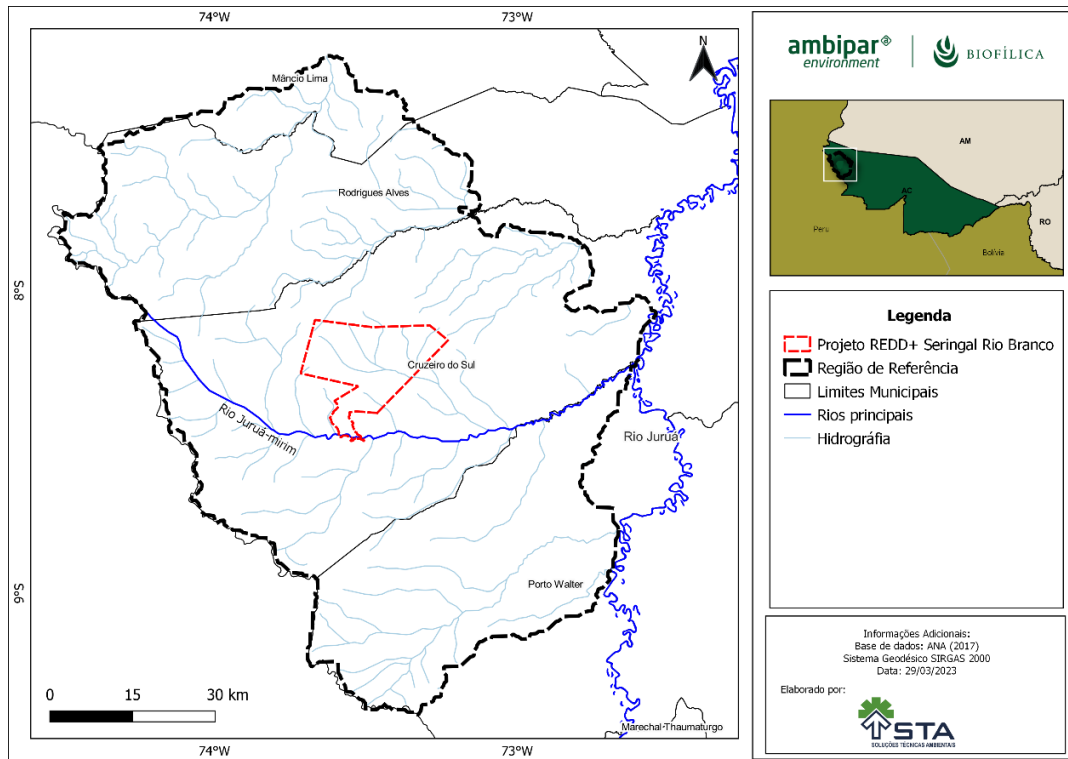


Figure 4 - Hydrography in the reference region of the Seringal Rio Branco REDD+ Project

Mineral Aspects:

To characterize the mineral resources of the project's reference region, a search was carried out on the Mining Geographic Information System (SIGMINE) website of the National Mining Agency (ANM).

From the data collection, it was possible to observe that within the Seringal Rio Branco REDD+ Project area there are no mineral resources with exploration potential. However, in the reference region, to the north, it was identified that there are the following mineral resources with exploration potential: Cassiterite (9805.61 ha); Amethyst (1709.97 ha); Diamond (11388.98 ha); Gold Ore (2664.46 ha).

Geological aspects:

In the reference region, according to the Environmental Information Database (BDiA) of the Brazilian Institute of Geography and Statistics (IBGE), the lithostratigraphic units are identified: Structural Subprovincials; Indiscriminate Cenozoic Coverage; Moa-Divisor; Solimões (Figure 6).

The predominant formation in the project area is that of Solimões. This formation predominates in almost the entire State of Acre, extending beyond the border into the Peruvian and Bolivian territories, with the north and south limited to Proterozoic terrain, and to the east, it thins over several pre-Cenozoic sedimentary formations. It is largely covered by sedimentation associated with the fluvial paleosystem of the Içá Formation and by Pleistocene detrital-laterite covers, being exposed in areas close to the valleys, especially those in the process of resuming erosion, in addition to presenting flat-parallel stratifications and tabular and channeled crosses of small, medium and large amplitude and

their sedimentation it was related to the general subsidence of the basin, with subsidence and uplift of varying intensity, resulting from the reactivation of old faults (Cavalcante, 2005)⁶.

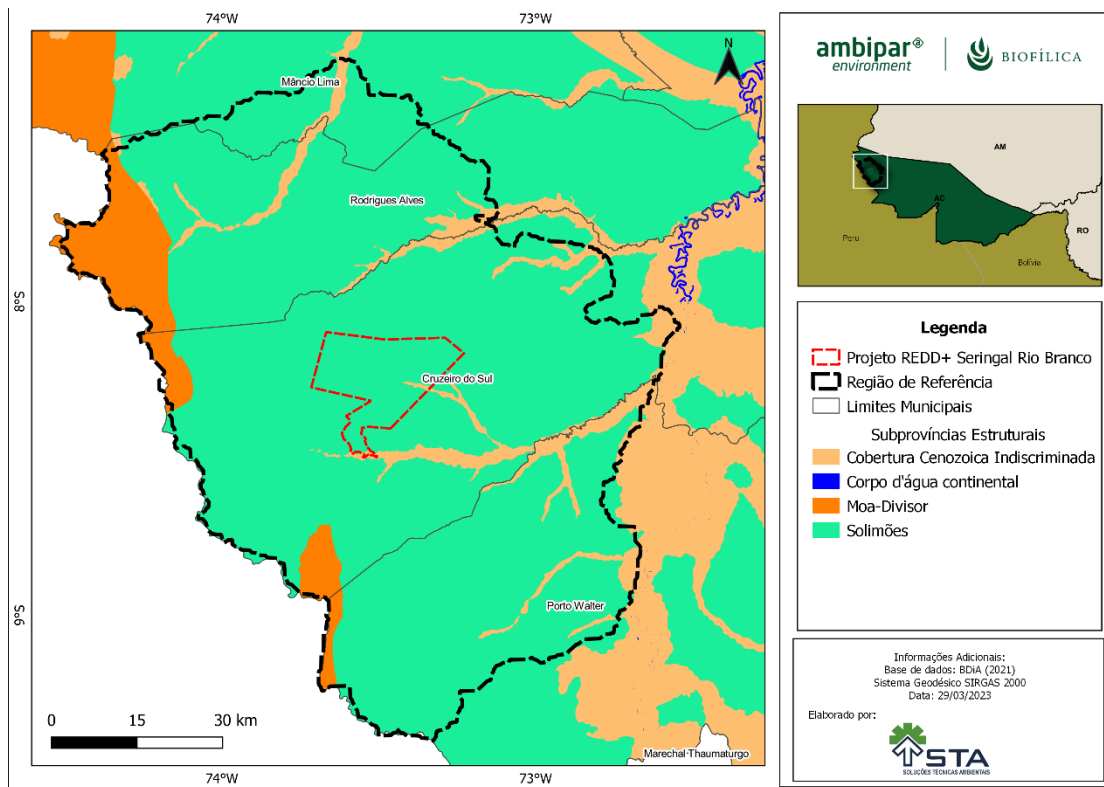


Figure 5 - Lithostratigraphic Units of the Seringal Rio Branco REDD+ Project reference area. Source: BDIA/IBGE (2021).

Geomorphological Aspects:

According to the BDIA (IBGE, 2021) and CPRM (2021) database, the reference region covers the following geomorphological units: Depression Marginal to Serra do Divisor; Tabular Surface of Cruzeiro do Sul; Juruá Depression; Amazon Plain; Flat. Residuals from Serra do Divisor (Figure).

The Seringal Rio Branco REDD+ Project is located in the Marginal Depression zone of Serra do Divisor. It is a unit with the following characteristics (Cavalcante, 2005): altitude varying from 230 to 300 m, chemical morphogenesis, and dendritic drainage pattern; constitutes a graben associated with the Batã Fault, with rearrangements through posttertiary pediplanation; the creation of drainages resulted in the dissection of the area; the dissection of this unit in the area takes place in a tabular and convex manner; dissected relief with convex tops, with gentle slopes, carved in lithologies of the Solimões Formation that originated mainly yellow Argisols in relation to argiluvic Plinthosols, haplic Cambisols

⁶ CAVALCANTE, L. M. Zoneamento geológico e geomorfológico entre Feijó e Mâncio Lima – Acre. Rio Branco: Embrapa Acre, 2005. 24 p. il. color. (Embrapa Acre. Documentos, 99). 2005.

and locally carbonate chromic Vertisols due to the presence of carbonate material in the geological formation.

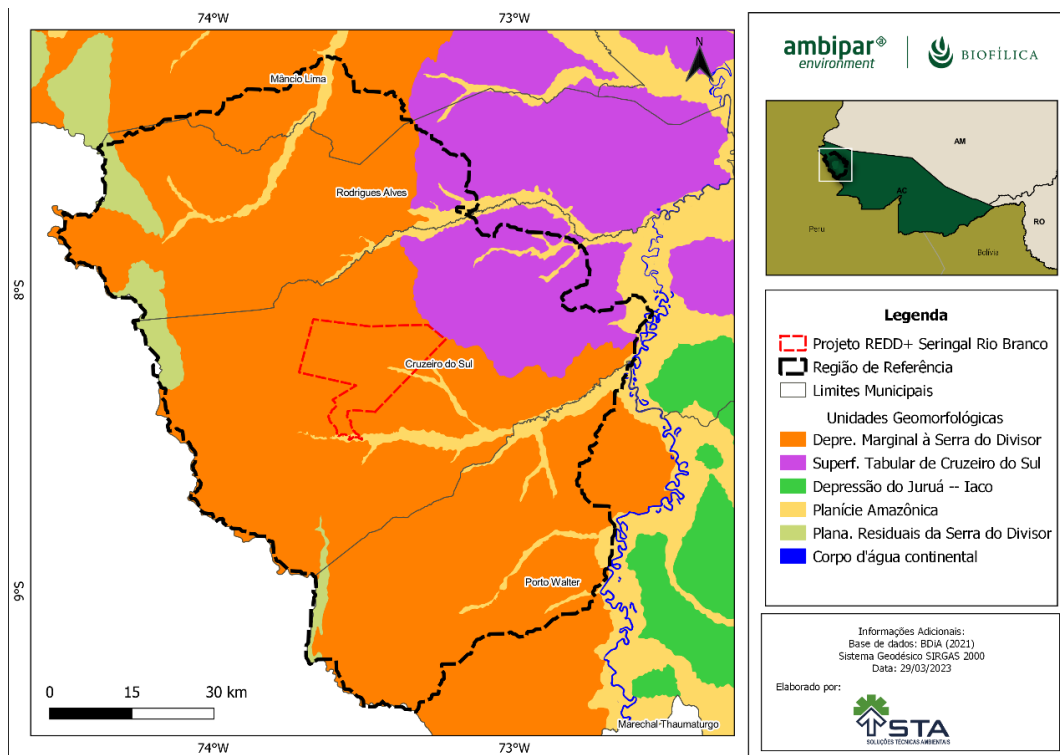


Figure 6 - Geomorphological Units of the Seringal Rio Branco REDD+ Project reference region. Source: BDIA/IBGE (2021).

Pedological Aspects:

Acre's soils have several orders, ranging from Oxisols (well-developed soils) to Fluvic Neossolos (young soils) (Do Amaral et al., 2021)⁷. Regarding pedological composition, the reference region covers areas with: Red-yellow Argisol; Haplic Gleisol; Chromic Luvisol; Yellow Argisol; Argiluvic Plinthosol. Red-Yellow argisols are predominant in the project area, as represented in Figure .

The Red-Yellow Argisols (PVA) have a clay accumulation horizon, textural B (Bt), with yellowish-red colors due to the presence of the mixture of iron oxides hematite and goethite, being deep and very deep soils, well-structured and well drained, resulting in horizons A, Bt; A, BA, Bt; A, E, Bt. Furthermore, the superficial A horizon of the moderate and prominent type is predominant, with a medium/clayey texture, with the possibility of having a medium/medium and medium/very clayey texture less

⁷ DO AMARAL, E. F.; LANI, J. L.; BARDALES, N. G.; DE ARAÚJO, E. A. Origem, formação e diversidade dos solos do Acre. Capítulo 2. 2021

frequently, in addition, these are soils whose natural fertility varies from low to very low, with a strongly acidic reaction and low activity clays (Da Silva; Neto, 2021)⁸.

This type of soil has the potential to be used for the cultivation of sugar cane, fruit cultivation, cassava cultivation, as well as passion fruit and yam, however, for good use, fertilization and liming must be applied, given the medium and low natural fertility.

In this sense, due to their low and medium natural fertility, in the Coastal Tabuleiros, such soils require correctives and fertilizers, in order to achieve good crop productivity, in addition to requiring the use of organic matter in the surface horizon, especially in soils with a sandy texture (Da Silva; Neto, 2021).

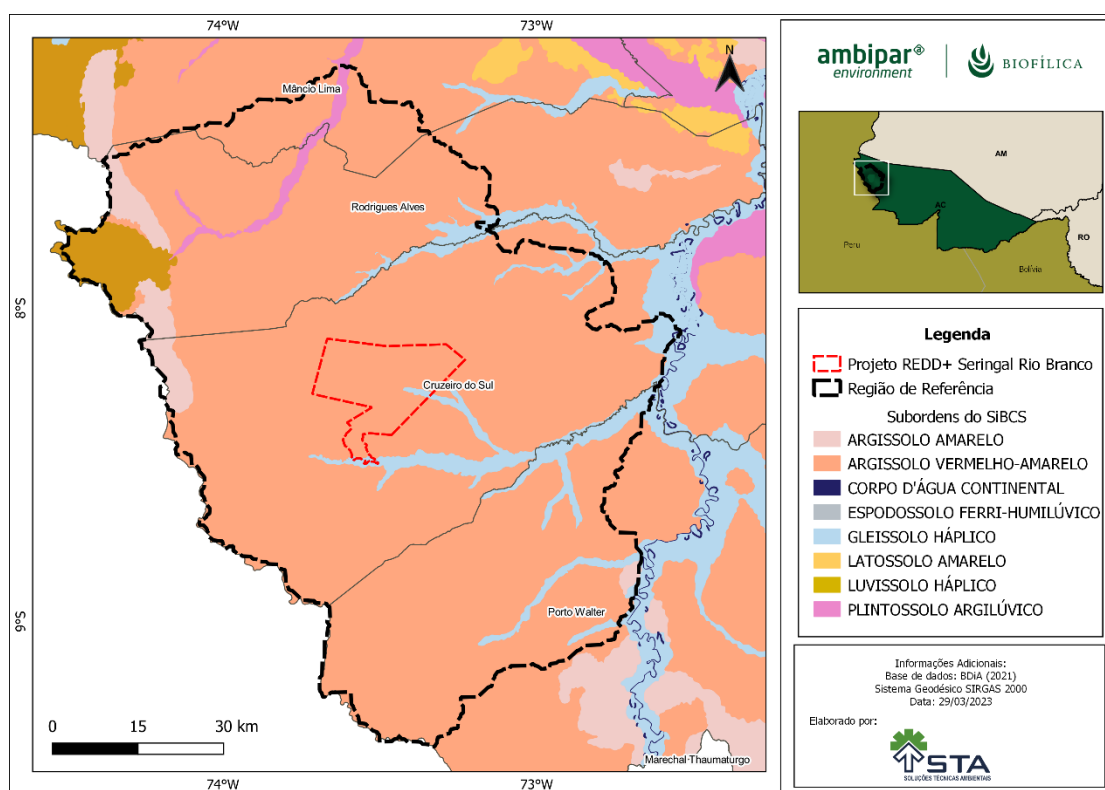


Figure 7 - Pedological characterization in the REDD+ Project reference region Seringal Rio Branco.
Source: BDIA/IBGE (2021).

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

⁸ DA SILVA, M. S. L.; NETO, M. B. de O. Argissolos Vermelho-Amarelos. Empresa Brasileira de Pesquisa Agropecuária – EMBRAPA. 2021.

Compliance with Laws, Statutes and other significant regulatory bodies for the Seringal Rio Branco REDD+ Project is related to the Non-Timber Forest Products Management activity. In the State of Acre, the project's activities are being licensed by the Brazilian Institute of the Environment and Renewable Natural Resources (IBAMA), therefore federal legislation must be applied. Subordinate to federal legislation, legislation applies at state level.

Regarding REDD+ activities, a history of initiatives can be noted despite the construction and negotiation of this concept through agreements and meetings at the United Nations Framework Convention on Climate Change. UNFCCC). In December 2015, Brazil's National Strategy for REDD+ (ENREDD+) was established by MMA Ordinance No. 370, a document that formalizes to Brazilian society and UNFCCC signatory countries how the Brazilian government has structured its efforts and intends to improve them by 2020, contributing to the mitigation of climate change by controlling deforestation and forest degradation, promoting forest recovery and promoting sustainable development. In this context, in Brazil, Decree No. 10,144 (of 11/28/2019) established the National Commission for REDD+ (CONAREDD+) in order to coordinate, monitor, monitor and review the National Strategy for REDD+ and guide the development of requirements for the access to payments for results of REDD+ policies and actions in the country. The following year, the internal regulations of CONAREDD+ were published, through an Ordinance (n° 544, of 10/26/2020).

At the same time, of a widely relevant nature, to date, Bill No. 572/2020 is under analysis, which “Establishes the national system for reducing emissions from deforestation and degradation, conservation, sustainable forest management, maintenance and increase in stocks of forest carbon (REDD+) and other measures”. The text is being processed in the Chamber of Deputies.

As for the carbon market, there is a Bill (PL n° 528 of 2021) being processed in the Chamber of Deputies that aims to establish the Brazilian Emissions Reduction Market (MBRE) and regulate the purchase and sale of carbon credits in the country arising from activities to Reduce Emissions from Deforestation and Forest Degradation, for example. The promotion of this voluntary carbon market is provided for in the Law that established the National Climate Change Policy (Law n° 12,187, of 12/29/2009).

After years of discussion and stagnation of PL n° 528 of 2021 in the National Congress, more recently, Decree n° 11,075 of 05/19/2022 was promulgated, which addresses the implementation of a regulated carbon credit market in Brazil through the creation of the National System for Reducing Greenhouse Gas Emissions (Sinare) and establishes procedures for the preparation of Sectoral Climate Change Mitigation Plans. In addition to these measures, the document also brings new concepts relating to methane credit, recording the carbon footprint of processes and activities, carbon from native vegetation, soil carbon and blue carbon.

Below, the main relevant legislation and regulations at federal and state levels are listed and detailed. Furthermore, a brief analysis was carried out of the international climate agreements that have been directing the creation and development of REDD+ initiatives around the world.

Federal Legislation

- Law No. 14.119, as of 01/13/2021: Institutes the National Policy on Payment for Environmental Services; and amends Laws No. 8.212, of July 24, 1991, 8.629, of February 25, 1993, and 6.015, of December 31, 1973, to adapt them to the new policy.
- Law No. 12.727, as of 12/17/2012: Provides for the protection of native vegetation; amends Laws No. 6.938, of August 31, 1981, 9.393, of December 19, 1996, and 11.428 of December 22, 2006; and repeals Laws No. 4.771, of September 15, 1965, and 7.754, of April 14, 1989, Provisional Measure No. 2.166-67, of August 24, 2001, item 22 of item II of art. 167 of Law No. 6.015, of December 31, 1973, and § 2 of art. 4 of Law No. 12.651, of May 25, 2012.
- Law No. 12.651, as of 05/25/2012: Provides for the protection of native vegetation; amends Laws No. 6.938, of August 31, 1981, 9.393, of December 19, 1996, and 11.428, of December 22, 2006; repeals Laws No. 4.771, of September 15, 1965, and 7.754, of April 14, 1989, and Provisional Measure No. 2.166-67, of August 24, 2001; and makes other provisions.
- Law No. 12.187, as of 12/29/2009: Establishes the National Policy on Climate Change – PNMC and other measures.
- Decree No. 11.075, as of 05/19/2022: Establishes the procedures for the preparation of Sectoral Plans for Climate Change Mitigation, establishes the National Greenhouse Gas Emission Reduction System and amends Decree No. 11.003, of March 21, 2022.
- Decree No. 10.144, as of 11/28/2019: Establishes the National Commission for the Reduction of Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Forest Management and Increase of Forest Carbon Stocks – REDD+.
- Decree No. 58.054, as of 03/23/1966: It promulgates the Convention for the protection of the flora, fauna and scenic beauties of the countries of America.
- Decree No. 2.661, as of 07/08/1998: Regulates the sole paragraph of art. 27 of Law No. 4.771, of September 15, 1965 (Forest Code), by establishing precautionary rules regarding the use of fire in agropastoral and forestry practices, and makes other provisions.
- Decree No. 5.975, as of 11/30/2006: Regulates art. 12, final part, 15, 16, 19, 20 and 21 of Law No. 4.771, of September 15, 1965, art. 4, item III, of Law No. 6.938, of August 31, 1981, art. 2 of Law No. 10.650, of April 16, 2003, amends and adds provisions to Decrees No. 6.514/08 and 3.420/00, and makes other provisions.
- Decree No. 10.936, as of 01/12/2022: It regulates Law No. 12.305, which establishes the National Policy for Solid Waste.
- CONAMA Resolution No. 16, as of 12/07/1989: Establishes the Integrated Program for Environmental Assessment and Control of the Legal Amazon.

- CONAMA Resolution No. 378, as of 10/19/2006: Defines the projects potentially causing national or regional environmental impact for the purposes of the provisions of item III, § 1, art. 19 of Law No. 4.771, of September 15, 1965, and other measures.
- CONAMA Resolution No. 379, as of 10/19/2006: Creates and regulates data and information system on forest management under the National Environmental System - SISNAMA.
- IBAMA Ordinance No. 218, as of 05/04/1989: Provides for the felling and exploitation of native forests and successor forest formations native to the Atlantic Forest, and makes other arrangements.
- IBAMA Ordinance No. 438, as of 08/09/1989: Amends the wording of Article 4 of Ordinance No. 218, of May 04, 1989.
- MMA Ordinance No. 103, as of 04/05/2006: Provides for the implementation of the Document of Forest Origin - DOF, and other measures.
- MMA Ordinance No. 253, as of 08/18/2006: Institutes, as of September 1, 2006, within the scope of the Brazilian Institute of Environment and Renewable Natural Resources - IBAMA, the Document of Forest Origin - DOF to replace the Authorization for Transportation of Forest Products - ATPF.
- Ordinance No. 1.896, of 12/09/2013: Amends Regulatory Standard No. 31.
- MMA Normative Instruction No. 1, as of 09/05/1996: Provides for the Mandatory Forest Replacement and the Integrated Forest Plan.
- MMA Normative Instruction No. 07, as of 04/27/1999: Provides for the authorization for deforestation in the States of the Legal Amazon.
- MMA Normative Instruction No. 02, as of 05/10/2001: Provides for the economic exploitation of forests, in rural properties located in the Legal Amazon, including the Legal Reserve areas and excepting those of permanent preservation established in the current legislation, which will be carried out through sustainable forest management practices of multiple use.
- MMA Normative Instruction No. 06, as of 12/15/2006: Provides for forest replacement and consumption of forest raw material, and other measures.
- IBAMA Normative Instruction No. 178, as of 06/23/2008: Defines the guidelines and procedures, by IBAMA, for the assessment and consent related to the issuance of authorizations for the suppression of forests and other forms of native vegetation in an area greater than two thousand hectares in rural properties located in the Legal Amazon and one thousand hectares in rural properties located in other regions of the country.
- Regulatory Standard No. 31, of 03/03/2005: Approves the Regulatory Standard for Occupational Safety and Health in Agriculture, Livestock, Forestry, Forest Exploration and Aquaculture.

State Legislation

- State Law no. 2,308, of October 22, 2010: "Creates the State System of Incentives for Environmental Services - SISA, the Incentive Program for Environmental Services - ISA Carbono and other Environmental Services and Ecosystem Products Programs of the State of Acre and provides other measures".
- State Law No. 1,426 of 12/27/2001: "Provides for the preservation and conservation of the State's forests, establishes the State System of Protected Natural Areas, creates the State Forestry Council and the State Forestry Fund and provides other measures." . The project is in compliance with this law as it presents a management plan for non-timber forest products and authorization for the use of forest raw materials from IBAMA and the Acre Environmental Institute (IMAC).
- State Law No. 3,883, of December 17, 2021: Provides for the State System of Protected Natural Areas
 - SEANP, which includes state, municipal and federal conservation units, indigenous lands, legal reserves (RL) and permanent preservation areas (APP). For RL and APP, SEANP aims to promote public policies and actions aimed at the sustainable use of natural resources, the formation of ecological corridors and other compatible measures. The project complies with this law as it includes activities aimed at the sustainable use of natural resources.
- State Law No. 1,548, of January 29, 2004: Amends art. 9th of Law no. 1,426, of December 27, 2001.
- State Law No. 2,836, of December 30, 2013: Amends Law No. 1,426, of December 27 of 2001, which provides for the preservation and conservation of the State's forests, establishes the State System of Protected Natural Areas, creates the State Forestry Council and the State Forestry Fund and provides other measures.
- State Law No. 3,595, of December 20, 2019: Amends provisions of Law No. 1,022, of January 21, 1992, Law No. 1,117, of January 26, 1994 and Law No. 1,426, of January 27 December 2001.
- Complementary State Law No. 300, of July 9, 2015: Amends Complementary Law No. 247, of February 17, 2012, which provides for the administrative structure of the Executive Branch; Law no. 2,308, of October 22, 2010, which creates the State System of Incentives for Environmental Services – SISA, the Incentive Program for Environmental Services – ISA Carbono and other Environmental Services and Ecosystem Products Programs of the State of Acre; Law no. 1,426, of December 27, 2001, which provides for the preservation and conservation of the State's forests, establishes the State System of Protected Natural Areas, creates the State Forest Council and the State Forest Fund, and provides other measures.
- State Law No. 1,460, of May 3, 2002. "Establishes the Support Program for Traditional Populations and Small Producers–PRÓ-FLORESTANIA, and provides other measures.
- State Law No. 1,904, of June 5, 2007. "Institutes Ecological – Economic Zoning of the State of Acre – ZEE.
- State Law No. 1,963, of December 4, 2007. "Provides for Plant Health Defense in the State of Acre.
- State Law No. 2,024, of October 20, 2008. "Creates the State Incentive Program for Family Forestry and Agroforestry Production.

- Decree No. 503, of April 6, 1999. “Establishes the State Ecological-Economic Zoning Program for the State of Acre, and provides other measures.
- Decree nº 8,452, of August 14, 2003. “Establishes the structure and composition of the State Forestry Council and regulates the Forestry Fund.
- Decree No. 3,414, of September 12, 2008. “Provides for forest replacement in the State of Acre due to the consumption of forest raw materials.
- Decree No. 3,415, of September 12, 2008. “Provides for the creation of the State Commission for Environmental Risk Management of Acre.
- Decree No. 5,507, of July 15, 2010. “Provides for the State Council for Sustainable Rural and Forestry Development - CDRFS.
- Resolution/CEMACT No. 001, of June 22, 2005. “Approves the terms of the IMAC/IBAMA Interinstitutional Ordinance No. 001 of 06/04/2005, which deals with the definition of minimum standards for the sustainable use of the unha-de-vine vine. cat.

Municipal Legislation – Cruzeiro do Sul

- Municipal Law No. 453, of October 7, 2006: Provides for the participatory Master Plan of the municipality of Cruzeiro do Sul and provides other measures”
- Municipal Law No. 457, of December 7, 2006: Provides for the Municipal Environmental Policy of the municipality of Cruzeiro do Sul, its purposes and formulation and application mechanisms, establishing the Municipal Environmental System and Changing the powers of SEMMA and COMDEMA, and provides other measures.

International Agreements

- FCCC/CP/2005/Misc.1: Reducing emissions from deforestation in developing countries: approaches to stimulate action. Submission from Parties. (Tradução: Reduzindo emissões de desmatamento em países em desenvolvimento: abordagem para estimular ação. Submissão das partes. COP 11, Montreal, 2005.)
- FCCC/CP/2007/6/add.1: Report of the Conference of the Parties on its thirteenth session, held in Bali from 3 to 15 December 2007. Addendum. Part two: Action taken by the Conference of the Parties at its thirteenth session. (Tradução: Relatório da Conferência das Partes sobre sua décima terceira sessão, ocorrida em Bali de 3 a 5 de dezembro de 2007. Addendum. Parte Dois: Ação tomada pela Conferência das Partes em sua décima terceira sessão ou “Action Bali Plan”. COP 13, Bali, 2007.)
- FCCC/CP/2009/Add.1: Report of the Conference of the Parties on its fifteenth session, held in Copenhagen from 7 to 19 December 2009. Addendum. Part Two: Action taken by the Conference of the Parties at its fifteenth session. (Tradução: Relatório da Conferência das Partes sobre sua décima quinta sessão, ocorrida em Copenhagen de 7 a 19 de dezembro de 2009. Addendum. Parte Dois:

Ação tomada pela Conferência das Partes na sua décima quinta sessão ou “Copenhagen Accord”. COP 15, Copenhagen, 2009.)

- FCCC/CP/2010/7/Add.1: Report of the Conference of the Parties on its sixteenth session, held in Cancun from 29 November to 10 December 2010. Addendum. Part Two: Action taken by the Conference of the Parties at its sixteenth session. (Tradução: Relatório da Conferência das Partes sobre sua décima sexta sessão, ocorrida em Cancun de 19 de novembro a 10 de dezembro de 2010. Addendum. Parte Dois: Ação tomada pela Conferência das Partes na sua décima sexta sessão ou “Cancun Agreement”. COP 16, Cancun, 2010.)

- FCCC/CP/2011/9/Add. 1: Report of the Conference of the Parties on its seventeenth session, held in Durban from 28 November to 11 December 2011. Addendum. Part Two: Action taken by the Conference of the Parties at its seventeenth session. (Tradução: Relatório da Conferência das Partes sobre sua décima sétima sessão, ocorrida em Durban de 28 de novembro a 11 de dezembro de 2011. Addendum. Parte Dois: Ação tomada pela Conferência das Partes em sua décima sétima sessão. COP 17, Durban, 2011.)

- FCCC/CP/2012/8/Add.1: Report of the Conference of the Parties on its eighteenth session, held in Doha from 26 November to 8 December 2012. Addendum. Part two: Action taken by the Conference of the Parties at its eighteenth session. (Tradução: Relatório de Conferência das Partes sobre sua décima oitava sessão, ocorrida em Doha de 26 de novembro a 8 de dezembro. Addendum. Parte Dois: Ação tomada pela Conferência das Partes em sua décima oitava sessão.)

- FCCC/CP/2013/Add.1: Warsaw Framework for REDD-plus, held in Warsaw, Poland, from 11 to 22 November 2013 (Tradução: Pacote de Varsóvia para REDD+, ocorrida em Varsóvia, Polônia, de 11 a 22 de Novembro de 2013), em especial as seguintes decisões:

- Decision9/CP.19: Work programme on results-based finance to progress the full implementation of the activities referred to in decision 1/CP. 16, paragraph 70. (Tradução: Programa de trabalho em financiamento baseados em resultados para o progresso da implementação completa das atividades referidas na decisão 1/CP. 16, parágrafo 70.)

- Decision10/CP.19: Coordination of support for the implementation of activities in relation to mitigation actions in the forest sector by developing countries, including institutional arrangements. (Tradução: Coordenação do suporte para a implementação de atividades relacionadas a ações de mitigação no setor florestal por países em desenvolvimento, incluindo arranjos institucionais.)

- Decision12/CP.19: The timing and the frequency of presentations of the summary of information on how all the safeguards referred to in decision1/CP.16, appendix I, are being addressed and respected. (Tradução: O tempo e a frequência na qual são apresentadas as informações resumidas de como todos os salvaguardas referidos na decisão1/CP.16, apêndice I, estão sendo abordadas e respeitadas.)

- Decision13/CP.19: Guidelines and procedures for the technical assessment of submissions from Parties on proposed forest reference emission levels and/or forest reference levels. (Tradução: Guia

e procedimentos para avaliação técnica das submissões das Partes em propostas de níveis de referência em emissões florestais e/ou níveis de referência florestal.)

- Decision14/CP.19: Modalities for measuring, reporting and verifying. (Tradução: Modalidades para medir, reportar e verificar.)

- Decision15/CP.19: Addressing the drivers of deforestation and forest degradation. (Abordagem dos vetores de desmatamento e degradação florestal.)

- FCCC/CP/2015/Add.1: Report of the Conference of the Parties on its twenty-first session, held in Paris from 30 November to 13 December 2015. Addendum. Part two: Action taken by the Conference of the Parties at its twenty-first session. (Tradução: Relatório de Conferência das Partes sobre sua vigésima primeira sessão, ocorrida em Paris de 30 de novembro a 13 de dezembro. Addendum. Parte Dois: Ação tomada pela Conferência das Partes em sua vigésima primeira sessão).

- FCCC/CP/2015 Paris Agreement: Global, legally-binding agreement that sets out a global framework to avoid dangerous climate change by limiting global warming to well below 2°C and pursuing efforts to limit it to 1.5°C. Entry into force on 4 November 2016.

- FCCC/CP/2016 Decisions adopted by the Conference of the Parties (COP): Especially decisions 1 (preparation into force of the Paris Agreement), 3 (Warsaw International Mechanism for Loss and Damage associated with Climate Change Impacts), 6 (National adaptation plans) and 7 (Long-term climate finance).

- FCCC/CP/2017, FCCC/CP/2018, FCCC/CP/2019 Decisions adopted by the COP: Especially decision 1 reporting on developments of the implementation of the Paris Agreement.

- Contribuição Nacionalmente Determinada – NDC brasileira encaminhada em setembro de 2015 à Convenção Quadro das Nações Unidas sobre Mudança do Clima para a mitigação, adaptação e meios de implementação, de maneira consistente com o propósito das contribuições de alcançar o objetivo último da Convenção, nos termos da decisão 1/CP.20, parágrafo 9.

- CITES, de 03/03/1973: “Convention on International Trade in Endangered Species of Wild Fauna and Flora”, assinada em Washington D.C. em 03 de março de 1973, alterado em Bonn em 22 de junho de 1979.

- Article 6 of the Paris Agreement (2021): Decision 1/CP.21 mandated the SBSTA to operationalize the provisions of this Article through recommending a set of decisions to the COP serving as the meeting of the Parties to the Paris Agreement at its first session. At COP26, the Parties to the Paris Agreement at its third session (CMA 3) adopted three main decisions related to Article 6: decision 2 (on Article 6.2), decision 3 (on Article 6.4) and decision 4 (on Article 6.8).

- Glasgow Leaders’ Declaration on Forests and Land Use (2021): Signatories (including Brazil) promise to reverse and end deforestation by 2030.

- Brazilian Nationally Determined Contribution (NDC): First Brazilian NDC submitted in September 2015 to the UN Framework Convention on Climate Change for mitigation, adaptation and means of

implementation, in a manner consistent with the purpose of contributions to achieve the ultimate objective of the Convention, pursuant to Decision 1/CP.20, paragraph 9. The updated Brazilian NDC was presented at the COP26 on December 8th, 2022.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The Seringal Rio Branco REDD+ Project has as one of its objectives to promote sustainable development in the region, with the joint actions of all interested parties, facilitated by Amazônia Agroindústria EIRELI, driving net benefits for the climate, biodiversity and local communities. Based on this support and in accordance with expected impacts, the project will contribute to the following UN sustainable development goals:

Goal 2. End hunger, achieve food security and improved nutrition, and promote sustainable agriculture		
2.4	By 2030, ensure sustainable food production systems and implement resilient agricultural practices that increase productivity and production, help maintain ecosystems, strengthen the capacity to adapt to climate change, extreme weather, droughts, floods and other disasters, and progressively improve land and soil quality	
The project combines the socioeconomic demands of communities with local opportunities aimed at more resilient economic activities, through “Fostering sustainable practices”. To this end, the project facilitates environmental education and community training actions, in association with different partners and extension workers, in order to promote knowledge about the importance of reconciling good production practices with the preservation and maintenance of natural resources. In addition to promoting perceptions about the relevance and advantages of sustainable production systems, the project also works to raise awareness among extractive families regarding the value of sustainable exploitation of non-timber forest resources for their survival, income generation, well-being and, consequently, keeping the forest standing.		
Goal 4. Ensure inclusive, equitable, quality education and promote lifelong learning opportunities for all		
4.4	By 2030, substantially increase the number of young people and adults who have relevant skills, including technical and professional skills, for employment, decent jobs and entrepreneurship.	
4.7	By 2030, ensure that all apprentices acquire the knowledge and skills necessary to promote sustainable development, including, but not limited to, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship, and appreciation of cultural diversity and of culture's contribution to sustainable development.	
The project enables and encourages access to education through technical courses and training focused on environmental and socioeconomic areas, especially on resilient agricultural production practices, sustainable extraction of non-timber forest products, education on hunting, fishing and preservation of endangered species. In addition, it provides specific training aimed at Seringal Rio Branco workers, especially those who belong to the patrimonial patrol, knowledge and guidelines on appropriate measures to be taken in cases of illegal activities in the area. To achieve this, it counts on the support and collaboration of specialized partners, in order to guarantee the		

effectiveness and engagement of interested parties. These training activities promoted by the project enable the strengthening of local governance, the consolidation of the feeling of belonging, access to information, better employment conditions and income diversification; mainly for extractivists and small rural producers, consequently resulting in the maintenance of the forest and its resources.		
Goal 12. Ensure sustainable production and consumption patterns		
12.2	By 2030, achieve sustainable management and efficient use of natural resources	
12.8	By 2030, ensure that people everywhere have relevant information and awareness for sustainable development and lifestyles in harmony with nature	
The project includes “Fostering sustainable practices” with actions aimed at identifying potential activities related to resilient subsistence agriculture, sustainable livestock farming and extractivism and low-impact forest management, according to the demand and profile of local communities. In this sense, the project works to disseminate knowledge, instructions and experiences focused on the efficient use of natural resources and environmental preservation; focusing on sustainable business chains through greater integration between interested parties, thus generating income, well-being and cultural identity for the fostered communities. The learning, engagement and pre-disposition of these families in activities to improve production and extractive practices increase project governance and help maintain forest cover and preserve its ecological aspects.		 
Goal 13. Take urgent action to combat climate change and its impacts		
13.2	Integrate climate change measures into national policies, strategies and planning	
13.3	Improve education, increase awareness and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning	
13.b	Promote mechanisms for building capacity for climate change-related planning and effective management in least developed countries, including with a focus on women, youth, local and marginalized communities	

In general, all activities developed by the project aim to take actions to combat climate change and its impacts and, consequently, reduce environmental degradation in the project area. In this sense, in addition to engaging stakeholders and strengthening governance, the project also promotes deforestation monitoring via satellite and in the field. Property surveillance within Seringal Rio Branco, equipped with geospatial information, is more effective and assertive in actions to prevent and combat illegal activities. For this to happen, the project works to improve the registration and dissemination of pressured and deforested sites in the project area, as well as investing in training workers to make appropriate decisions in cases of identification of irregular activities. As a result, reducing greenhouse gas emissions, directly collaborating with the Brazilian goal of reducing emissions.		
Goal 15. Protect, restore and promote the sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, halt and reverse land degradation and halt the loss of biodiversity		
15.1	By 2020, ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains and drylands, in accordance with obligations arising from international agreements	
15.2	By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests, and substantially increase afforestation and reforestation globally	
15.5	Take urgent and significant measures to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2020, protect and prevent the extinction of threatened species	
15.7	Take urgent action to end illegal hunting and trafficking of protected flora and fauna species and address both the demand and supply of illegal products of life	
15.a	Mobilize and significantly increase, from all sources, financial resources for the conservation and sustainable use of biodiversity and ecosystems	
15.b	Mobilize significant resources from all sources and at all levels to finance sustainable forest management and provide appropriate incentives to developing countries to promote sustainable forest management, including for conservation and reforestation	
15.c	Strengthen global support for efforts to combat illegal hunting and trafficking of protected species, including by increasing the capacity of local communities to pursue sustainable livelihood opportunities	



Considering the relevance of Seringal Rio Branco for biodiversity; the project focuses efforts on long-term monitoring of ecological indicators associated with practices arising from REDD+ conservation activities on populations and ecosystems, inside and outside the property limits. In this way, the project aims to minimize habitat loss, landscape changes, overexploitation of species and climate change. To this end, it seeks to engage, involve and raise awareness of all interested parties regarding the importance of fauna and flora biodiversity in the provision of ecosystem services, maintaining landscape connectivity, controlling environmental degradation and limiting the excessive use of natural resources. Environmental education lectures on hunting and fishing; training on sustainable livelihoods and workshops and activities on local ecological aspects are aimed mainly at workers and community members in the surrounding area, including extractivists and riverside dwellers; that move around the property and use forest resources. This participatory involvement, access to knowledge and results, and knowledge of good production practices; promote the strengthening of local governance and the inhibition of irregular actions; ensuring positive impacts on biodiversity in the project area.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Under development

1.19.2 Commercially Sensitive Information

Some information required by the VCS and/or CCB standard is considered confidential or commercially sensitive and cannot be publicly disclosed by Project proponents. This information was fully provided to the audit team during the validation process attached to this document but was not included in the public version. Below is a list of information that was made available:

- Land documents and legal status.
- Proponents' financial statements;
- Project financial performance spreadsheet (budget) and other related documents.
- Agreements and contracts signed between the parties involved.
- Inventories and other diagnoses.

1.19.3 Further Information

Under development

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Under development.

Stakeholder Identification	<i>Describe the process(es) used to identify stakeholders likely impacted by the project. List the stakeholders identified.</i>
Legal or customary tenure/access rights	<i>Describe any legal or customary tenure/access rights to territories and resources, including collective and conflicting rights, held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.</i>
Stakeholder diversity and changes over time	<i>Describe the social, economic, and cultural diversity within stakeholder groups, the differences and interactions between the stakeholder groups, and any changes in the make-up of each group over time.</i>
Expected changes in well-being	<i>Describe the expected changes in well-being and other stakeholder characteristics relative to the baseline scenario, including changes to ecosystem services identified as important to stakeholders;</i>
Location of stakeholders	<i>Describe the location of stakeholders, IPs, LCs, and customary rights holders, and areas outside the project area that are predicted to be impacted by the project.</i>

Location of resources	<i>Describe the location of territories and resources which stakeholders own or to which they have customary access.</i>
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2.1.2 Stakeholder Consultation and Ongoing Communication

Under development.

Date of stakeholder consultation	<i>DD-Month-YYYY</i>
Stakeholder engagement process	<i>Describe the process to engage stakeholders in a culturally appropriate manner (e.g., dates of announcements or meetings, language and gender sensitivity). Describe the process or methods used to document the outcomes.</i>
Consultation outcome	<i>Summarize the discussion around consent to project design and implementation, risks, costs and benefits of the project, all relevant laws and regulations covering workers' rights in the host country, the discussion of FPIC and the VCS validation and verification process.</i>
Ongoing communication	<i>Describe the mechanisms for ongoing communication with stakeholders.</i>
Stakeholder input	<i>Describe how due account was taken of all input received during the consultation. Include details on any updates to the project design or justify why updates were not necessary or appropriate.</i>

2.1.3 Free Prior and Informed Consent

Under development.

Obtaining consent	<i>Describe and demonstrate how consent to implement the project activities was obtained from those concerned, including IPs, LCs, and customary rights holders, and a transparent agreement was reached. Describe any ongoing or unresolved conflicts and demonstrate that the project does not exacerbate nor influence the outcomes of unresolved conflicts.</i>
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Outcome of FPIC	<i>Describe the outcome of the FPIC process, the transparent agreement, and the information disclosed prior to establishing a transparent agreement with those concerned, IPs, LCs, and customary rights holders. Provide assurance that the project has not encroached on land, relocated people without consent, and forced physical or economic displacement.</i>
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2.1.4 Grievance Redress Procedure

Under development.

Development process	<i>Describe the process used to develop the grievance redress procedure including processes for receiving, hearing, responding and attempting to resolve grievances within a reasonable time period, taking into account culturally appropriate conflict resolution methods.</i>
Grievance redress procedure	<i>Describe the grievance redress procedures developed with stakeholders.</i>

2.1.5 Public Comments

Under development.

Comments received	Actions taken
<i>Summary of comment received</i>	<i>Provide a summary of actions taken and any project design updates or justify why updates were not necessary or appropriate.</i>
...	

2.2 Risks to Stakeholders and the Environment

Under development.

Risks identified	Mitigation or preventative measure taken
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Risks to stakeholder participation		
Working conditions		
Safety of women and girls		
Safety of minority and marginalized groups, including children		
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)		

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Under development.

Discrimination and sexual harassment	Through its Codes of Conduct, the project team seeks to establish and promote ethical values and principles of social and environmental responsibility, of general acceptance and applicable to all its direct and indirect collaborators, also including suppliers, service providers, partners, between others. The Code of Conduct prohibits any embarrassment, abuse of power or harassment of any kind, such as sexual, economic or moral. These guidelines are taken to the project's stakeholders, who, as partners, are obliged to follow the same premises established in the Code of Conduct.
Management experience	Proponente: Plínio Ribeiro – Executive Director Plínio Ribeiro holds a degree in Business Administration from the Instituto de Ensino e Pesquisa - INSPER and a master's degree in public administration and Environment from Columbia University and the Earth Institute (USA). He has participated in several conservation projects on the lower Negro River, through the Institute of Ecological Research – IPÊ since 2005 and was one of the producers of the documentary "Retorno à Amazônia", by Jean Michel Cousteau. He has been working at Biofílca

Ambipar Environmental Investments S.A since 2008, where he has led Projects, Operations and Business Management. He is co-founder and CEO of Biofílica Ambipar Environmental Investments S.A.

Cláudio Pádua – Scientific Director

Cláudio Pádua holds a degree in Business Administration and Biology, a Master's in Latin American Studies and a PhD in Ecology from the University of Florida in Gainesville (USA). Retired professor at the University of Brasilia, Pádua is currently dean of the Superior School of Conservation and Sustainability and vice-president of the Institute of Ecological Research (IPÊ). He is also Senior Associate Researcher at the Center for Environment and Conservation Studies at Columbia University (USA) and Director of International Conservation at the Wildlife Trust Alliance, as well as a consultant for the Brazilian Biodiversity Fund (FUNBIO) and WWF Brazil. Pádua represents Brazil before the International Advisory Group (IAG) of the G7 Pilot Program. In 2003, together with his wife, Suzana Pádua, he was named "Hero of the Planet" by Time magazine for his activities in favor of biodiversity conservation. Between 1997 and 2007, he won six conservation awards, three national and three internationals. Pádua has published two books and more than 30 articles in national and international scientific journals. Since 2008 he has directed Biofílica's Ambipar Environmental Investments S.A involvement and scientific production as Scientific Director and advisor.

Paula Conde – Administrative Manager

Paula Conde has a degree in Business Administration from PUC of São Luís and a postgraduate degree in Accounting and Financial Administration from FAAP. She has extensive experience, most of it in one of the largest media and education groups in Latin America - Editora Abril, where she worked with Financial Control and Reporting, Treasury, Accounting and Financial Reconciliation, Accounts Payable and Receivable and Royalties. At Biofílica Ambipar Environmental Investments S.A, she is responsible for administrative and financial activities, logistical support for the team and projects. She has been at Biofílica Ambipar Environmental Investments S.A since 2015, being

responsible for administrative and financial activities, specializing in REDD+ Carbon Market, intermediation, billing of these segments, logistical support to the team and projects.

Soraya Pires – Director of Operations

Agronomist, graduated from Esalq/USP with specialization in strategic management and finance from FGV, with 15 years of experience in management and development of agribusiness businesses and career developed in large multinational companies in the sugar-energy sector (Adecoagro, BP and BP Bunge). Since 2022, she has been an executive director in the carbon market, carrying out activities related to business development, operational management, financial modeling, strategic planning and AFOLU project management.

Caio Gallego – Intelligence Manager

Caio Gallego is a Forestry Engineer graduated from the University of São Paulo (USP/ESALQ). Specialist in geoprocessing and remote sensing focused on the area of environmental conservation, mapping and analysis of changes in land use. He has knowledge focused on Sustainable Forest Management, environmental modeling and the use of alternative GIS for forestry and agribusiness. He has advanced knowledge in the use of GIS software and analysis of changes in land use and land cover such as ArcGIS, QuantumGIS and DinâmicaEGO. Caio Gallego has more than 8 years of experience working with the development and implementation of AFOLU projects for voluntary carbon markets. He worked with project development, with a main focus on implementing one of the first REDD+ projects within the VCS program in the Brazilian Amazon. His main expertise has always been focused on the technical development of projects, having been responsible for more than 10 validation/verification processes. In more than 10 projects in which he was directly responsible for development, he worked on the application of socio-environmental safeguards, ensuring the generation of net positive impacts on climate, community and biodiversity, in addition to ensuring the maintenance of exceptional criteria applicable to each of these scopes.

Laion Pazian – Commercial Manager

Laion Pazian is an Economist from the University of São Paulo (USP/ESALQ) and an MBA in Commercial Management from Fundação Getúlio Vargas. He has been at Biofílica Ambipar Environmental Investments S.A since 2016, and manages the commercial team of carbon credits, key-accounts, Biofílica's Ambipar Environmental Investments S.A commercial policy and strategy. In addition, he monitors and directs the analysis of carbon market intelligence, being responsible for the area's pricing and planning policy.

Ricardo Cordeiro – Communication Coordinator

Ricardo Cordeiro is an advertiser, On and Offline art director with more than 10 years of experience, he has worked in digital agencies, trade and live marketing. Experience in UX, planning and digital strategies. Specialization in Digital Marketing and Web Project Management. At Biofílica Ambipar Environmental Investments S.A, where he has been working since 2020, he acts as communication coordinator, responsible for digital marketing, branding and institutional communication actions.

Márcio Sales – Statistical Modeling Specialist

Márcio Sales is a statistician, graduated from the Federal University of Pará, Master in Geography from the University of California, Santa Barbara and PhD student in Production Ecology and Resource Conservation from the University of Wageningen, in the Netherlands. He specializes in data analysis and conducts research in geostatistical modeling of processes distributed in space and time. He works at Biofílica Ambipar Environmental Investments S.A in the production of projections of GHG emissions from future deforestation for the baselines of the projects and in the monitoring of deforestation by satellite.

Shaxahmary de Mori – Project Coordinator

Shaxahmary de Mori is a Forest Engineer, graduated from the University of São Paulo (USP/ESALQ) and specialist in Project Management. In his last professional experiences, he had an interface with themes of natural resource conservation, climate change, payments for environmental services, restoration and forest hydrology. Shaxahmary has approximately three years of experience in developing and

monitoring REDD+ projects and currently operates as a carbon project coordinator.

Amanda Rocha Fiallos - Project Analyst

Amanda Fiallos is a Forest Engineer, graduated from the University of São Paulo (USP/ESALQ). She has experience in forest restoration and conservation projects, environmental education, geoprocessing and remote sensing applied to the planning areas of planted forests. She has been working in Biophilic since May 2022, producing technical reports and reporting activities of the AFOLU Projects, and evaluating reports and products delivered by suppliers regarding the socio-environmental diagnoses made.

Bianca de Oliveira – Project Analyst

Bianca de Oliveira is a Forest Engineer, graduated from the Federal Rural University of Rio de Janeiro, and a specialist in Integrated Environmental Management from the State University of Rio de Janeiro. Professional who has expertise in environmental licensing, forest restoration, climate change and conservation. She has been carrying out activities for the preparation, development and monitoring of AFOLU projects since 2023.

Raphael Ramiro – FP&A Specialist

Administrator graduated from Universidade Estadual Paulista (UNESP) and Postgraduate in Corporate Finance from Universidade Federal de São Carlos (UFSCAR). Professional with more than 12 years of experience in analysis and evaluation of projects from an economic-financial point of view. At Biofílica Ambipar Environmental Investments S.A, he performs activities related to the Evaluation and Financial Modeling of Carbon Projects (AR, REDD+, ALM, Blue Carbon), Business Valuation (M&A), Analysis of Financial Additionality - Carbon Projects, Management / Board of Directors Presentations, Financial Performance Analysis / KPIs.

Marco Antonio Martins – Geoprocessing Analyst

Marco Antonio Martins is a Geographer graduated from the University of São Paulo (FFLCH-USP) and a master's student in the areas of environmental planning and remote sensing, working with mapping priority areas for forest restoration on farms. During his graduation, he worked with geoprocessing and remote sensing projects applied to

forest monitoring, territorial planning, mapping of coastal habitats and restored areas in conservation units. Since 2021, he has been working on AFOLU projects, developing geographic intelligence analyzes for forest conservation, forest corridor restoration and low-carbon agriculture projects.

Franciane Almeida - Geoprocessing Analyst

Franciane Almeida is a Forestry Engineer, graduated from the Federal University of Lavras (UFLA). She has experience in the area of geoprocessing and remote sensing applied to the areas of forest planning and harvesting, recovery of degraded areas and land regularization. She also has experience in AFOLU projects, mainly in monitoring forest cover of REDD+ projects based on satellite images and available official deforestation databases, such as PRODES.

Proponents: Amazônia Agroindústria EIRELI

James Cameli - Amazônia Agroindústria EIRELI

Location: Rio Branco, AC - Brazil James Cameli, successful businessman and recognized throughout the western Brazilian Amazon. Owner of a conglomerate of 5 companies, operating for over 30 years in the sectors of agriculture, telecommunications, construction, industrialization and processing of forestry products and environmental services. He is a supporter of environmental conservation policies and projects, and is convinced that the "standing forest" is one of the region's greatest assets and, therefore, represents a strategic asset for the social and economic development of Brazil and its contribution to change current global climate conditions.

Scarlett Siqueira do Valle - Amazônia Agroindústria EIRELI (Legal adviser)

Location: São Paulo, São Paulo, Brazil

Scarlett Siqueira do Valle is an independent lawyer (401449 OAB/SP), graduated in Law from the Centro Universitário de Brasília - UNICEUB and pursuing a PhD in Communication and Semiotics from the Pontifical Catholic University of São Paulo - PUC-SP. She has experience in the areas of Startup Law; Digital Law; LGPD and Business Law. As a corporate lawyer, she has experience in environmental conservation projects.

	Veriton Viana da Costa - Amazônia Agroindústria EIRELI (Technical Advisor and Local Support) Location: Cruzeiro do Sul, AC - Brazil Veriton Costa is an agronomist and agricultural technician, graduated from the Federal University of Acre (UFAC). He has experience in Financial Management, Registration of traditional populations, Analysis of physical and chemical soils, Geoprocessing (ArcGis), People Management, Project Preparation, Report Preparation (Power BI and Excel), Inspection in Agricultural and Forestry Defense, Lectures and Courses. At Amazônia Agroindústria EIRELI he works as Advisor and Local Support.
Gender equity in labor and work	Through its Codes of Conduct, the project team seeks to establish and promote ethical values and principles of social and environmental responsibility, of general acceptance and applicable to all its direct and indirect collaborators, including also suppliers, service providers, partners, between others. As established in the Code of Conduct, all employment decisions are made with respect for diversity, and no discriminatory or segregational conduct should be practiced, whether based on race, color, nationality, sex, religion, age, sexual orientation, origin, disabilities, beliefs policies, role or any other personal characteristics. These guidelines are taken to the project's stakeholders, who, as partners, are obliged to follow the same premises established in the Code of Conduct.
Human trafficking, forced labor, and child labor	Through its Codes of Conduct, the project team seeks to establish and promote ethical values and principles of social and environmental responsibility, of general acceptance and applicable to all its direct and indirect collaborators, including also suppliers, service providers, partners, between others. The Code of Conduct prohibits any form of child labor and any form of forced labor or labor in conditions similar to slavery. These guidelines are taken to the project's stakeholders, who, as partners, are obliged to follow the same premises established in the Code of Conduct.

2.3.2 Human Rights

Under development.

2.3.3 Indigenous Peoples and Cultural Heritage

Under development.

2.3.4 Property Rights

Under development.

Rights to territories and resources	<i>Describe any legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights, held by stakeholders.</i>
Respect for property rights	<i>Describe the measures implemented to protect and preserve the property rights of IPs, LCs, and customary rights holders.</i>

2.3.5 Benefit Sharing

Under development.

Process used to design the benefit sharing plan	<i>Describe the process used to develop the benefit-sharing agreement with the affected stakeholder groups.</i>
Summary of the benefit sharing plan	<i>Describe the benefit-sharing agreement. Where affected stakeholder groups wish to keep elements of the plan private, provide the full arrangement as a commercially sensitive document. The project proponent shall demonstrate that the community wishes to keep this information private.</i>
Approval and dissemination of benefit sharing plan	<i>Demonstrate that the benefit-sharing agreement was agreed up on by the affected stakeholder groups, and that the agreement was shared in a culturally appropriate manner. Demonstrate that the agreement is readily accessible should stakeholders wish to review the agreement.</i>

2.4 Ecosystem Health

Under development.

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems		
Soil degradation and soil erosion		
Water consumption and stress		
Usage of fertilizers		

2.4.1 Rare, Threatened, and Endangered species

Under development.

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☐ No

If yes, list such species and habitats in the table below and provide evidence that the project will not adversely impact these areas.

Species and habitat	<i>Demonstrate that the project will not adversely impact habitats for rare, threatened, or endangered species.</i>
...	...

2.4.2 Introduction of species

Under development.

Species introduced	Classification	Justification for use	Adverse effects and mitigation

Where invasive species exist in the project area, list such species in the table below and demonstrate that the project activity will not allow the species to thrive.

Existing invasive species	Mitigation measures to prevent spread or continued existence of invasive species
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2.4.3 Ecosystem conversion

Not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The Seringal Rio Branco REDD+ Project utilizes the "VM0007 REDD+ Methodology Framework version 1.6" (hereafter referred to as REDD-MF). Two activities were deemed eligible as part of this project: 1) avoiding planned deforestation; and 2) avoiding unplanned deforestation. Therefore, modules related to these two activities are necessary.

This project is eligible as a project to avoid planned deforestation because forested lands are expected to be converted into non-forested lands, with legal permission for such conversion, as permitted by the Brazilian Forest Code, allowing 20% of the area of properties located in the Amazon to be converted to other land uses. Additionally, there is the risk of invasions and illegal extraction of natural resources due to anthropogenic pressure surrounding the project area, making the project also eligible to avoid unplanned deforestation.

The "VM0007 REDD+ Methodology Framework version 1.6" (REDD-MF) is referred to as a modular methodology as it provides the generic functionality of the methodology, framing pre-defined modules and tools that perform specific functions. Together with the modules and tools it employs, it constitutes a complete methodology for the baseline and monitoring of REDD+ projects, suitable for projects with activities reducing emissions from both planned deforestation (APD, Avoided Planned Deforestation) and unplanned deforestation (AUDD, Avoided Unplanned Deforestation and Degradation).

The modules and tools supporting REDD-MF are available in the virtual environment of Verra or the Clean Development Mechanism (CDM).

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF),	1.6
Module	BL-PL	VMD0006 Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL).	1.3
Module	BL-UP	VMD0007 Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation (BL-UP).	3.3
Module	CP-AB	VMD0001 Estimation of carbon stocks in the above and belowground biomass in live tree and non-tree pools.	1.1
	CP-D	VMD0002 Estimation of carbon stocks in the dead wood pool (CP-D)	1.0
	CP-W ⁹	VMD0005 Estimation of carbon stocks in the long-term wood products pool (CP-W), v1.1	1.1
Module	M-REDD	VMD0015 Methods for monitoring of greenhouse gas emissions and removals in REDD project activities (M-REDD).	2.2
Module	E-BPB	VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E- BPB)	1.2
Module	X-STR	VMD0016 Methods for stratification of the project area (X-STR).	1.2
Module	X-UNC	VMD0017 Estimation of uncertainty for REDD+ project activities (X-UNC).	2.2

⁹ O módulo CP-W não foi utilizado na parte do Inventário Florestal realizado na área do Projeto REDD+ Seringal Rio Branco, contudo ele foi considerado no Plano de Monitoramento do Projeto

Tool	T-ADD	CDM Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities (T-ADD)	1
Tool	T-BAR	VCS AFOLU Non-Permanence Risk Tool (T-BAR)	4
Tool	T-SIG	CDM Tool for testing significance of GHG emissions in A/R CDM project activities.	1

3.2 Applicability of Methodology

The methodology approved by the VCS, code VM0007, was used for the Seringal Rio Branco REDD+ Project and is applicable according to the applicability criteria specified.

Methodology ID	Applicability condition	Justification of compliance
VM0007 - REDD+ Methodology Framework (REDD+MF)	The land in the project area has been qualified as forest for at least 10 years prior to the project start date. The project area may include forested wetlands (such as floodplain forests), provided they do not grow on peat. Peat should be defined as organic soils with at least 65% organic matter and a minimum thickness of 50 cm3. If the project area includes forested wetlands growing on peat (e.g., peat swamp forests), this methodology is not applicable. Deforestation and forest degradation in the project area baseline fall into one or more of the following categories: Unplanned	The project area meets this condition, with complete forest cover demonstrated in the historical records of the past 10 years. As demonstrated below, there are no organic soils (or peatlands) in the project area. Additionally, soil carbon has been conservatively excluded, as indicated in section 5.1.5; therefore, the project is not subject to VCS WRC requirements. Baseline deforestation in the project area falls into the category of planned deforestation, as the project proponents are the agents of deforestation; and into the

	deforestation (VCS AUDD category); Planned deforestation (VCS APD category); Degradation through wood extraction for fuel (wood and charcoal production) (VCS AUDD category). Baselines will be renewed every 10 years from the project start date. All land areas registered under the CDM or any other carbon trading scheme (both voluntary and compliance-oriented) must be transparently reported and excluded from the project area. If the land is not being converted to an alternative use but natural regeneration is allowed (i.e., temporarily without stock), this framework should not be used. Leakage prevention activities should not include: Agricultural lands flooded to increase production (e.g., rice in wetlands); or intensification of livestock production through the use of "feedlots".	category of unplanned deforestation as there is the possibility of population growth in the project vicinity and hence illegal encroachments onto project lands. The baseline will be renewed every 10 years starting from the year 2023. This is the first registration of the area in a carbon trading program; the area will also be registered under the CCB. Baseline forest deforestation is followed by the establishment of managed pastures, which prevents forest regrowth. Leakage prevention activities do not include flooding agricultural lands or creating feedlots.
VMD0006 Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest	The prevention of planned deforestation/degradation activities is applicable under the following conditions:	The activity of preventing planned deforestation and degradation is applicable, given that the conversion of forested areas to deforested condition is legally permitted

degradation and planned wetland degradation (BL-PL)	Where the conversion of forested lands to a deforested condition must be legally permitted. When unsustainable wood collection is occurring within the project boundaries prior to the project, BL-DFW and LK-DFW modules should be used to identify potential leaks.	up to the limit of 20% in relation to the total area of the property, as stipulated in Article 12 of the Forest Code, Law 12,651 of 2012, described in section 2.5.7 National and Local Laws (G5.6). No wood extraction is carried out within the project area.
VMD0007 Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation (BL-UP).	The activities to prevent unplanned deforestation are applicable under the following conditions: The key agents of deforestation should: (i) clear land for tree harvesting, settlements, agricultural production (farmer), or livestock or aquaculture, when such deforestation for agricultural or livestock or aquaculture production does not represent industrial agriculture or large-scale aquaculture activities; (ii) do not have legally documented and undisputed	The activity of preventing unplanned deforestation is applicable. Deforestation agents in the baseline scenario clear the land for agriculture or livestock as the final land use. These agents do not have the legal right to use or clear the land. In the baseline scenario of this activity, post-deforestation land use does not constitute reforestation (i.e., agriculture or pasture are the final land uses). No wood extraction is carried out within the project area.

	right to deforest land for these purposes; and (iii) be residents in the deforestation reference region or be migrants. Under any other condition, this methodology should not be used. If, in the reference scenario to prevent unplanned deforestation project activities, post-deforestation land use constitutes reforestation, this methodology cannot be used. Where unsustainable wood collection was occurring before the project within the project boundaries, BL-DFW and LK-DFW modules must be used to determine potential leaks.	
VMD0001 Estimation of carbon stocks in the above and belowground biomass in live tree and non-tree pools (CP-AB)	The module allows for ex-ante estimation of carbon stocks in above and below-ground tree and non-tree biomass in the baseline scenario (pre- and post-deforestation stocks), as well as in the project case, and for ex-post estimation of changes in above and below-ground tree biomass carbon stocks in the project case. The module applies to all types of forests and age classes. The inclusion of above-ground tree biomass	This module applies to all types of forests and age classes. The inclusion of above-ground tree biomass reservoir as part of the project boundaries is mandatory according to the REDD-MF module; thus, biomass of trees with DBH > 10cm will be considered. Above-ground non-tree biomass will be considered starting from the biomass of palms (with DBH > 10cm); as

	reservoir as part of the project is mandatory. Non-tree above-ground biomass should be included as part of the project boundary if the following applicability criteria are met: Non-tree above-ground biomass stocks are higher in the baseline than in the project scenario; and, Non-tree above-ground biomass is considered significant (using the T-SIG module).	demonstrated, this biomass is significant according to the T-SIG tool. Subterranean biomass (tree and non-tree) will also be considered in carbon stock estimates, which is optional according to methodology requirements; as demonstrated, this biomass is significant according to the T-SIG tool.
VMD0002 Estimation of carbon stocks in the dead wood pool (CP-D)	This module is applicable to all types of forests and age classes. This module is applicable if the deadwood reservoir is included as part of the project.	This module is applicable to all types of forests and age classes. This reservoir is included when it represents a significant component of forest biomass.
VMD0005 Estimation of carbon stocks in the long-term wood products pool (CP-W)	This module allows for the ex-ante estimation of carbon stocks in the long-term timber products pool in the baseline scenario. The carbon stocks addressed here are those that remain in the wood products after 100 years; most emissions associated with harvesting, processing, and wood waste, as well as any removal of products, occur within this time period, and this module employs the simplifying assumption that the	This module is applicable in all cases where wood is harvested for conversion into wood products for commercial markets, for all types of forests and age classes. This module is applicable in the baseline, since the wood products pool is included as part of the project boundaries, according to the applicability criteria of the REDD-MF structural module, specifically:

	remaining proportion after 100 years is effectively "permanent."	Wood extraction occurs before or during the deforestation process (in the REDD+ Seringal Rio Branco project, wood extraction should occur during, as it will be one of the project activities) and the wood is destined for commercial markets; The wood products pool is considered significant.
VMD0015 Methods for monitoring of greenhouse gas emissions and removals in REDD project activities (M-REDD)	Emissions from logging activities can be omitted if it can be demonstrated that emissions are minimal through the T-SIG. If emissions from logging activities are not omitted, logging can only occur within forest management areas that possess and maintain Forest Stewardship Council (FSC) certification for the years when selective logging occurs. Logging operations can only conduct selective extraction that maintains land cover meeting the definition of forest within the project boundaries. All trees cut for logging during logging operations must have a diameter at breast height (DBH) greater than 30 cm. During cutting operations, only the trunk of the felled	Emissions from logging activities have not been omitted, as the commercial wood harvesting activity is expected to occur during the project, even though such activity does not currently exist.

	tree can be removed. The tree crown must remain within the forest area. Logging practices cannot include the stacking and/or burning of wood cuts. The volume of harvested wood must be measured and monitored.	
VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E-BPB)	This module provides a step-by-step approach to estimate greenhouse gas emissions from the burning of biomass and peat. This module is applicable to REDD project activities with emissions from biomass burning and REDD-WRC project activities with emissions from biomass and/or peat burning. This module is also applicable to RWE and ARR-RWE project activities with emissions from peat burning.	In the baseline scenario, fire is used for land clearing, resulting in greenhouse gas emissions. When used in the baseline, accounting should occur both in the baseline and project scenarios, and both within the project area and in the leakage belt. When post-fire incidents occur in areas coinciding with deforested or degraded areas in the baseline, the module should be used to account for greenhouse gas emissions.
VMD0016 Methods for stratification of the project area (X-STR)	This module provides guidance on how to stratify the project area into discrete and relatively homogeneous units to improve the accuracy and precision of carbon stock and carbon stock change estimates.	It was verified that there were no discrete biomass clusters that significantly differed from the average, so only one stratum, "Open Ombrophilous Forests," was considered within the project area. The post-deforestation (conversion) scenario will not be stratified; instead, average carbon stock values

		for land uses after deforestation were applied, following the guidelines of the BL-UP and BL-PL modules.
VMD0017 Estimation of uncertainty for REDD+ project activities (X-UNC)	The module allows for estimating uncertainty in baseline estimates and in project sequestration, emissions, and leakage estimates. The module focuses on the following sources of uncertainty: Determination of deforestation and degradation rates. Uncertainty associated with carbon stock estimates and changes in carbon reservoirs. Uncertainty in project emissions assessment. When an uncertainty value is unknown or cannot be easily calculated, the project must justify the use of a decidedly conservative value, and a 0% uncertainty may be applied to this component.	The module is mandatory when using the REDD-MF methodology.
CDM Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities (T-ADD)	Identical or similar AFOLU activities to the proposed project activity on-site within the proposed project boundaries, whether conducted with or without registration as a VCS AFOLU project, will not result in the violation of any applicable	To the best of our knowledge, Biofíllica may use this module to establish additionality to the project, as we believe the project is compliant with all relevant laws, statutes, and regulatory frameworks.

	laws, even if the law is not enforced. The use of this tool to determine additionality requires that the baseline methodology provides a stepwise approach justifying the determination of the most plausible baseline scenario. Project proponents proposing new baseline methodologies must ensure consistency between determining a baseline scenario and determining the additionality of a project activity.	The approach to defining the baseline scenario using this tool will not be described in this report but should be outlined in the PDD.
VCS AFOLU Non-Permanence Risk Tool (T-BAR)	This tool provides procedures for conducting non-permanence risk analysis and determining the necessary buffer for Agriculture, Forestry, and Other Land Use (AFOLU) projects. The tool sets forth requirements for project proponents and validation/verification bodies to assess risk and determine the appropriate risk classification.	The module is mandatory when using the REDD-MF methodology. The approach to defining risks using this tool will not be described in this report but should be outlined in the Project Design Document (PD).
CDM Tool for testing significance of GHG emissions in A/R CDM project activities (T-SIG)	This tool is used to verify the significance of reservoirs.	Recommended for all REDD+ projects.

3.3 Project Boundary

Under development.

Source		Gas	Included?	Justification/Explanation
Baseline	Source 1	CO ₂		
		CH ₄		
		N ₂ O		
		Other		
	Source 2	CO ₂		
		CH ₄		
		N ₂ O		
		Other		
Project	Source 1	CO ₂		
		CH ₄		
		N ₂ O		
		Other		
	Source 2	CO ₂		
		CH ₄		
		N ₂ O		
		Other		

3.4 Baseline Scenario

Under development.

3.5 Additionality

Regulatory

Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes☒ No

3.5.1 Additionality Methods

The additionality of the Seringal Rio Branco REDD+ Project was analysed using the tool "Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities (T-ADD) - Version 1", indicated by methodology VM0007 - REDD+ Methodology Framework (REDD+ MF), v1.6 to identify alternative land use scenarios and demonstrate the additionality of the project.

The tool used considers the following criteria to assess its applicability to the project:

- Forest conservation within the proposed project boundary, with or without a record of VCS REDD project activities, must not lead to a violation of any applicable law.
- The tool is not applicable to small-scale afforestation and reforestation project activities.

Considering the Seringal Rio Branco REDD+ Project, it can be said that it fulfils the first requirement, given that the project complies with local, regional and national legislation. As for the second applicability criterion, considering that the project is large-scale and not related to reforestation activities, it is understood that it fulfils the requirement.

In this way, the additionality analysis of the project can be carried out, using the tool mentioned above, in both the AUD and APD components.

APD

STEP 0. Preliminary identification based on the start date of the VCS AFOLU project activity

At this stage, according to the T-ADD tool (Version 1), if the project proponents report that the VCS REDD project activity has a start date after 31 December 1999, then the proponents must provide the following evidence:

- Evidence that the project start date was after 31 December 1999

- Evidence that the incentive of the planned sale of VCU was considered in the decision to proceed with the project activity, preferably based on official data and legal documentation, available to third parties at the time of or before the start of the project activity.

Insert information on the start date of the project, consider documentation that proves any request to the environmental agency to suppress vegetation (evidence item 1).

For item 2, this can be proved by the contract between the project proponents and biophilica.

STEP 1. Identify alternative land use scenarios for the proposed AFOLU project activity.

Sub-step 1a: Identify credible alternative land use scenarios for the proposed VCS AFOLU project activity.

SCENARIO 1 - Continuation of pre-project land use

The first scenario considers the continuation of activities observed prior to project implementation. This scenario is based only on monitoring activities in the area. In this scenario, considering the extension of the area, many riverside communities located in the surroundings use this land for their family farming and agriculture. One of the most common activities among the riverside communities around the project area is family farming, with the cultivation of manioc, the production of flour and other derived products.

According to the socio-economic diagnosis of the Seringal Rio Branco REDD+ Project, it is also possible to highlight the projected growth of farming practices and predatory hunting and fishing in the communities surrounding the project area, which have a significant impact on the area.

SCENARIO 2 - Land use project activity within the project boundary being carried out without registration of VCS AFOLU project activity

In the second scenario, it is considered that there is no economic activity of any kind being implemented in the area. This results in maintenance and surveillance costs for the area, without any economic return, even with legal authorisation to do so.

In this scenario, due to the costs of conserving the forest areas, it is not considered to be financially sustainable in the long term, considering the territorial extension of the area, and much less attractive for the owners, without the revenue generated by carbon credits.

SCENARIO 3 - Small-scale beef cattle farming

Scenario 3 envisages the use of land for beef cattle. In view of the legal permission to use this area and authorisation from the competent body to suppress vegetation, this scenario becomes attractive to landowners, with no VCS AFOLU project and consequently no generation of credits.

Considering the local reality and the trend in the region's municipalities, the use of land for livestock tends to bring greater economic benefits to landowners, not only through the sale of livestock, but also through the commercialisation of timber from cutting down the area.

Sub-step 1b. - Consistency of credible land use scenarios with applicable laws and regulations

Scenarios 1 and 2 comply with legal requirements, given that the scenarios do not take into account any type of economic activity, it is considered that the property's documentation is the only aspect of major relevance to the area and this complies with legal requirements.

For scenario 3, deforestation and livestock farming in the area are legally authorised under the Brazilian Forest Code, which allows 20% of the property's area to be used, with the remaining 80% set aside for conservation as a Legal Reserve. The 20% area can be used for various purposes, subject to a prior licence from the relevant environmental agency.

The three scenarios therefore comply with local and national legislation and regulations.

Sub-step 1c. - Selection of the base scenario

Based on the analyses highlighted above, it is possible to consider scenario 3 as the most pertinent to be selected as the base scenario for the Seringal Rio Branco REDD+ Project. This choice is justified by the predominance of livestock farming in the region, the projection of future growth, and also by the fact that it is legally feasible to clear the area, considering that this is the area of 20% of the property.

STEP 2 - Barrier Analysis

Sub-step 2a - Identification of barriers that would prevent the implementation of at least one of the alternative land use scenarios

3.6 Methodology Deviations

Under development.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Under development.

4.2 Project Emissions

Under development.

4.3 Leakage Emissions

Under development.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

For projects required to assess permanence risk:

i) Provide the requested information using the table below:

State the non-permanence risk rating (%)	12%
Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☒ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e, the Long-term Average (LTA).	N/A
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☒ No If no, provide justification.

State, in tCO ₂ e, the expected total GHG benefit to date.	
Is the number of GHG credits issued below the LTA?	☐ Yes ☒ No If no, provide justification.

Table . Ex ante estimate of net anthropogenic greenhouse gas emissions reductions (Δ REDDt) and Verified Carbon Units (VCUt) (Table 36 of Methodology VM0007).

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
18-Mar-2023 to 17-Mar-2024	227.092	22.709	22.709	27.592	27.592	27.592	154.082
18-Mar-2024 to 17-Mar-2025	229.436	22.944	22.944	27.877	27.877	55.468	155.672
18-Mar-2025 to 17-Mar-2026	231.781	23.178	23.178	28.161	28.161	83.630	157.263
18-Mar-2026 to 17-Mar-2027	240.292	21.626	21.626	29.520	29.520	113.149	167.520
18-Mar-2027 to 17-Mar-2028	242.700	21.843	21.843	29.816	29.816	142.965	169.199
18-Mar-2028 to 17-Mar-2029	245.109	22.060	22.060	30.112	30.112	173.077	170.877
18-Mar-2029 to 17-Mar-2030	247.517	22.277	22.277	30.407	30.407	203.484	172.556
18-Mar-2030 to 17-Mar-2031	256.092	20.487	20.487	31.807	31.807	235.291	183.311
18-Mar-2031 to 17-Mar-2032	258.564	20.685	20.685	32.114	32.114	267.404	185.080

18-Mar-2032 to 17-Mar- 2033	261.036	20.883	20.883	32.421	32.421	299.825	186.849
18-Mar-2033 to 17-Mar- 2034	31.308	2.505	2.505	3.889	3.889	303.714	22.411
18-Mar-2034 to 17-Mar- 2035	35.259	2.468	2.468	4.427	4.427	308.140	25.896
18-Mar-2035 to 17-Mar- 2036	33.105	2.317	2.317	4.156	4.156	312.297	24.314
18-Mar-2036 to 17-Mar- 2037	30.813	2.157	2.157	3.869	3.869	316.165	22.631
18-Mar-2037 to 17-Mar- 2038	34.763	2.086	2.086	4.411	4.411	320.577	26.180
18-Mar-2038 to 17-Mar- 2039	32.610	1.957	1.957	4.138	4.138	324.715	24.559
18-Mar-2039 to 17-Mar- 2040	30.457	1.827	1.827	3.865	3.865	328.580	22.937
18-Mar-2040 to 17-Mar- 2041	28.164	1.690	1.690	3.574	3.574	332.154	21.211
18-Mar-2041 to 17-Mar- 2042	32.115	1.606	1.606	4.119	4.119	336.273	24.784
18-Mar-2042 to 17-Mar- 2043	29.961	1.498	1.498	3.843	3.843	340.115	23.123
18-Mar-2043 to 17-Mar- 2044	32.915	1.646	1.646	4.221	4.221	344.337	25.402
18-Mar-2044 to 17-Mar- 2045	32.967	1.648	1.648	4.228	4.228	348.565	25.443

18-Mar-2045 to 17-Mar- 2046	39.262	1.570	1.570	5.088	5.088	353.653	31.033
18-Mar-2046 to 17-Mar- 2047	39.453	1.578	1.578	5.113	5.113	358.766	31.184
18-Mar-2047 to 17-Mar- 2048	39.505	1.580	1.580	5.120	5.120	363.886	31.225
18-Mar-2048 to 17-Mar- 2049	39.633	1.585	1.585	5.136	5.136	369.022	31.326
18-Mar-2049 to 17-Mar- 2050	45.927	1.378	1.378	6.014	6.014	375.036	37.157
18-Mar-2050 to 17-Mar- 2051	46.118	1.384	1.384	6.039	6.039	381.076	37.312
18-Mar-2051 to 17-Mar- 2052	46.171	1.385	1.385	6.046	6.046	387.122	37.354
18-Mar-2052 to 17-Mar- 2053	52.465	1.049	1.049	6.941	6.941	394.063	43.426
18-Mar-2053 to 17-Mar- 2054	52.656	1.053	1.053	6.966	6.966	401.029	43.584
18-Mar-2054 to 17-Mar- 2055	52.847	1.057	1.057	6.992	6.992	408.021	43.742
18-Mar-2055 to 17-Mar- 2056	52.761	1.055	1.055	6.980	6.980	415.001	43.670
18-Mar-2056 to 17-Mar- 2057	59.194	592	592	7.911	7.911	422.913	50.099
18-Mar-2057 to 17-Mar- 2058	59.385	594	594	7.937	7.937	430.849	50.261

18-Mar-2058 to 17-Mar-2059	59.576	596	596	7.962	7.962	438.812	50.422
18-Mar-2059 to 17-Mar-2060	59.629	596	596	7.969	7.969	446.781	50.467
18-Mar-2060 to 17-Mar-2061	65.923	0	0	8.900	8.900	455.681	57.024
18-Mar-2061 to 17-Mar-2062	66.114	0	0	8.925	8.925	464.606	57.189
18-Mar-2062 to 17-Mar-2063	64.918	0	0	8.764	8.764	473.370	56.154
Total	3.765.595	259.150	259.150	473.370	473.370	12.857.203	2.773.926

5 MONITORING

5.1 Data and Parameters Available at Validation

Under development.

Data / Parameter	
Data unit	Indicate the unit of measure
Description	Provide a brief description of the data/parameter
Source of data	Indicate the source(s) of data
Value applied	Provide the value applied
Justification of choice of data or description of	Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the

measurement methods and procedures applied	<i>responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix.</i>
Purpose of data	<i>Indicate one of the following:</i> • <i>Determination of baseline scenario (AFOLU projects only)</i> • <i>Calculation of baseline emissions</i> • <i>Calculation of project emissions</i> • <i>Calculation of leakage</i>
Comments	<i>Provide any additional comments</i>

5.2 Data and Parameters Monitored

Under development.

Data / Parameter	
Data unit	<i>Indicate the unit of measure</i>
Description	<i>Provide a brief description of the data/parameter</i>
Source of data	<i>Indicate the source(s) of data</i>
Description of measurement methods and procedures to be applied	<i>Specify the measurement methods and procedures, any standards or protocols to be followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests).</i>
Frequency of monitoring/recording	<i>Specify measurement and recording frequency</i>
Value applied	<i>Provide an estimated value for the data/parameter</i>
Monitoring equipment	<i>Identify equipment used to monitor the data/parameter including type, accuracy class, and serial number of equipment, as appropriate.</i>
QA/QC procedures to be applied	<i>Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.</i>

Purpose of data	<i>Indicate one of the following:</i> • <i>Calculation of baseline emissions</i> • <i>Calculation of project emissions</i> • <i>Calculation of leakage</i>
Calculation method	<i>Where relevant, provide the calculation method, including any equations, used to establish the data/parameter.</i>
Comments	<i>Provide any additional comments</i>

5.3 Monitoring Plan

Under development.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.